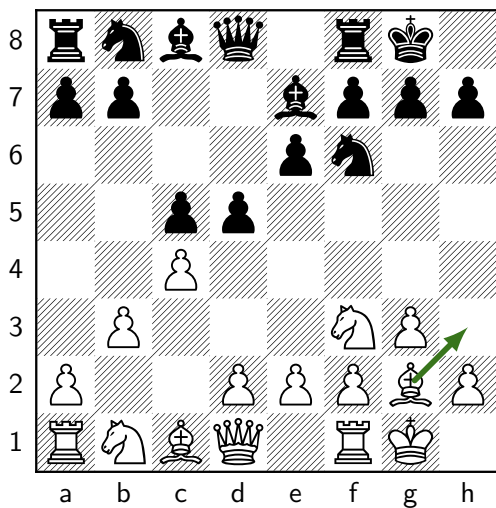


Melanie-BOTS (CCRL engines) — Reti Opening

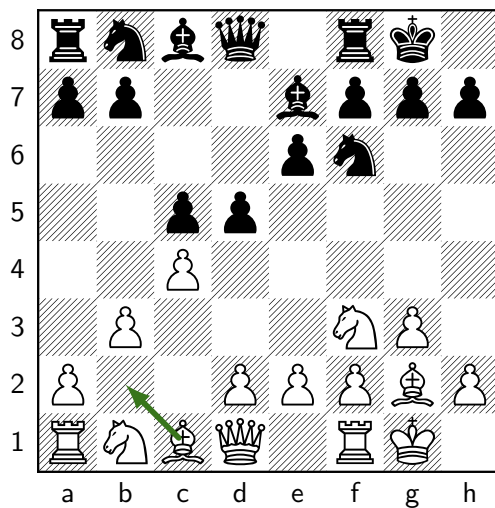
Melanie (Deep_Vision)

Abstract

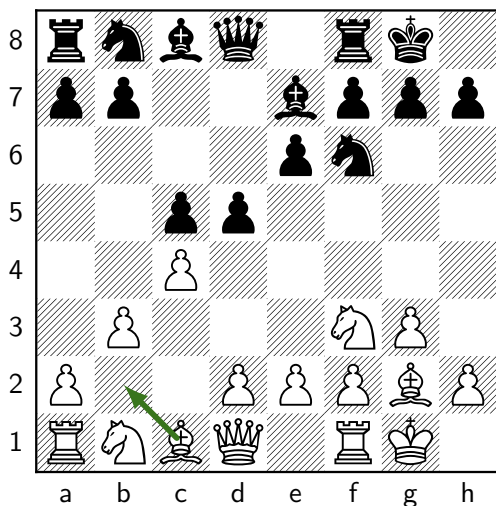
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie’s move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen’s role) are called out, as no material/eval engine explains a move that way.



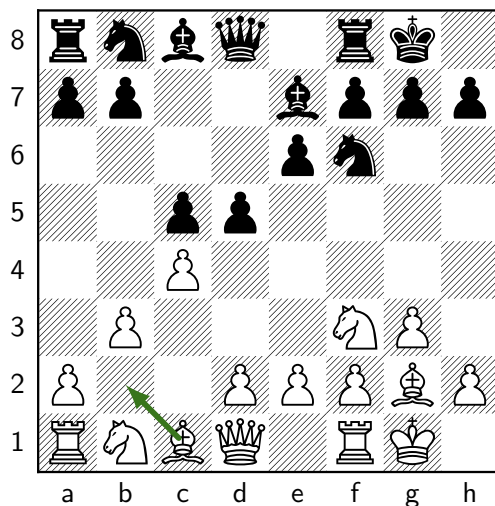
(a)



(b)

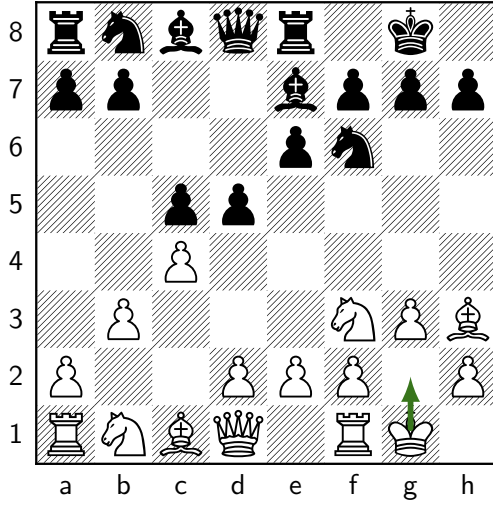


(c)

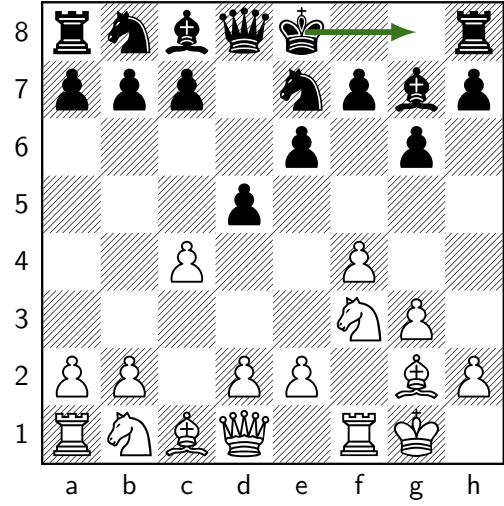


(d)

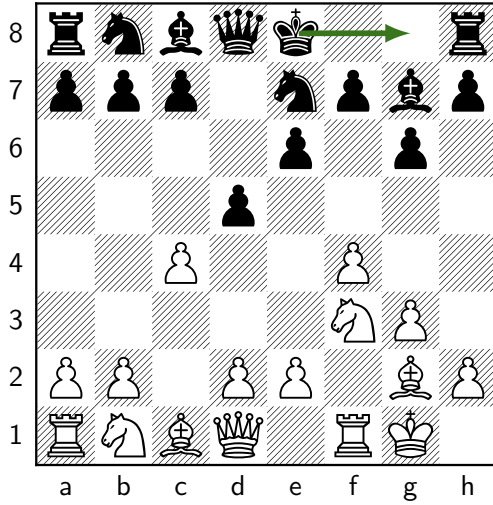
Figure 1: **Move 7, White: Bh3.** (a) Our position; Melanie plays Bh3 (g2h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1254` (ply 12) [structural distance $d=0$, QAP obj=53]: the master played Bb2; under μ our Bg2 plays the role of that game’s Bc1; cross-type roles: our Qd1 as their Ra1, our Rf1 as their Nb1; transferred move g2h3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1255` (ply 12) [structural distance $d=0$, QAP obj=53]: the master played Bb2; under μ our Bg2 plays the role of that game’s Bc1; cross-type roles: our Qd1 as their Ra1, our Rf1 as their Nb1; transferred move g2h3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#1446` (ply 12) [structural distance $d=0$, QAP obj=53]: the master played Bb2; under μ our Bg2 plays the role of that game’s Bc1; cross-type roles: our Qd1 as their Ra1, our Rf1 as their Nb1; transferred move g2h3 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Bh3 (summed QAP objective 316, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



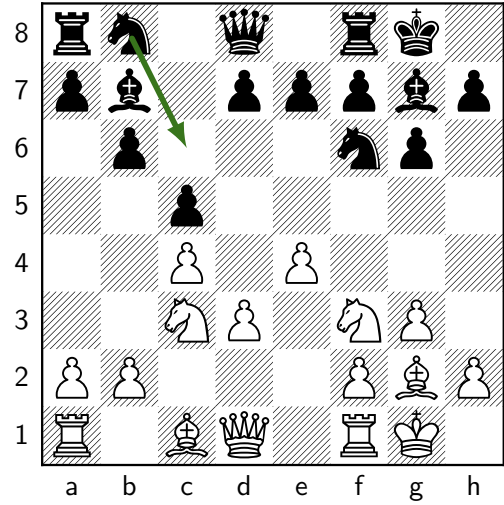
(a)



(b)

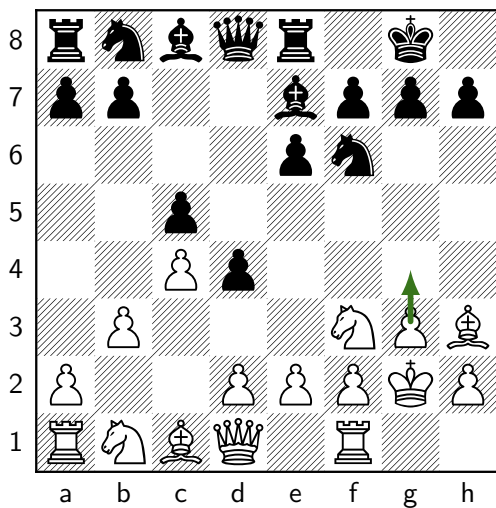


(c)

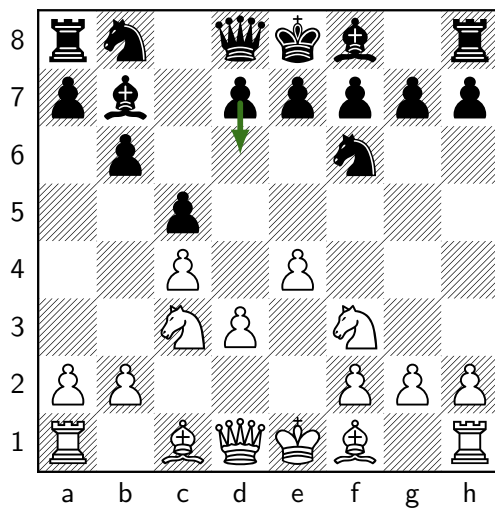


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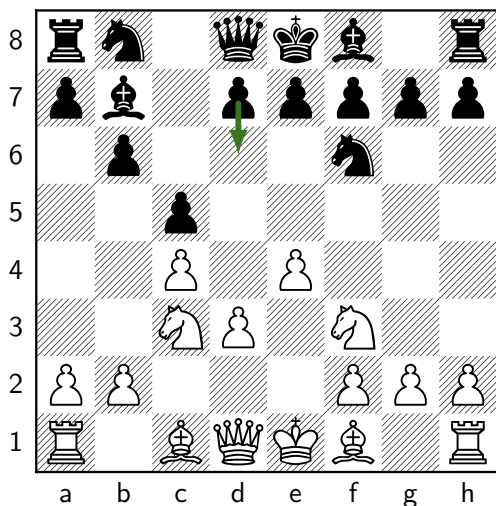
Figure 2: **Move 8, White: Kg2.** (a) Our position; Melanie plays Kg2 (g1g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2955` (ply 11) [structural distance $d=50$, QAP obj=58]: the master played 0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Pg3* as their *Ra8*, our *Ra1* as their *Nb8*; transferred move g1g2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2956` (ply 11) [structural distance $d=50$, QAP obj=58]: the master played 0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Pg3* as their *Ra8*, our *Ra1* as their *Nb8*; transferred move g1g2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1225` (ply 15) [structural distance $d=49$, QAP obj=41]: the master played Nc6; under μ our *Nb1* plays the role of that game’s *Nb1*; cross-type roles: our *pb7* as their *Ra8*, our *Bc1* as their *Pe7*; transferred move b1c3. Synthesis: 2 retrieved analogs transferred to Kg2 (summed QAP objective 116, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



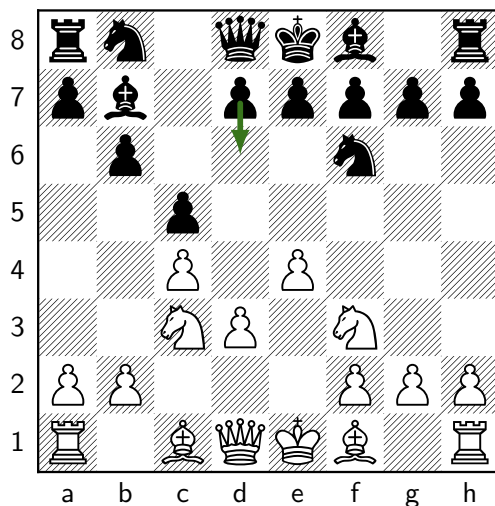
(a)



(b)

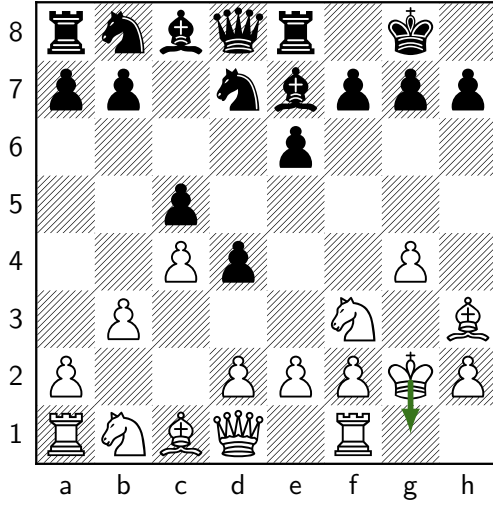


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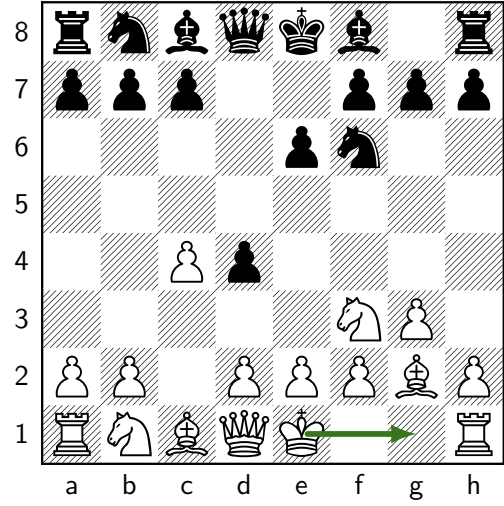


(d)

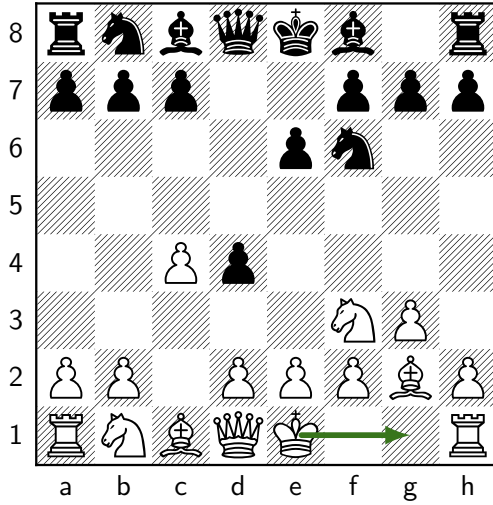
Figure 3: **Move 9, White: g4.** (a) Our position; Melanie plays g4 (g3g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top.p0001#494` (ply 9) [structural distance $d=60$, QAP obj=47]: the master played d6; under μ our $Pg3$ plays the role of that game’s $Pd2$; cross-type roles: our $Qd1$ as their $Ra8$, our $Pf2$ as their $Nb8$; transferred move g3g4 (**endorses this move**). (c) Retrieved from `ccr1_top.p0001#495` (ply 9) [structural distance $d=60$, QAP obj=47]: the master played d6; under μ our $Pg3$ plays the role of that game’s $Pd2$; cross-type roles: our $Qd1$ as their $Ra8$, our $Pf2$ as their $Nb8$; transferred move g3g4 (**endorses this move**). (d) Retrieved from `ccr1_top.p0001#888` (ply 9) [structural distance $d=60$, QAP obj=47]: the master played d6; under μ our $Pg3$ plays the role of that game’s $Pd2$; cross-type roles: our $Qd1$ as their $Ra8$, our $Pf2$ as their $Nb8$; transferred move g3g4 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to g4 (summed QAP objective 281, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



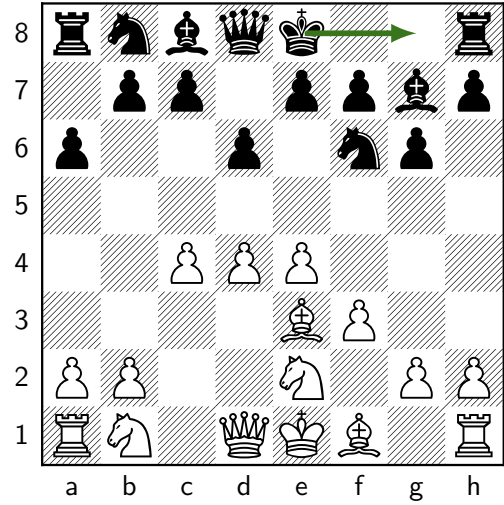
(a)



(b)

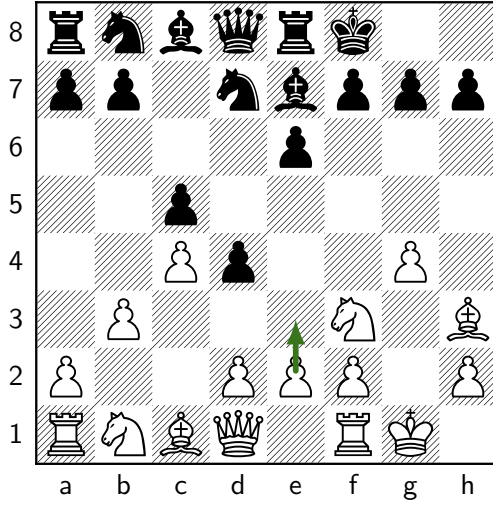


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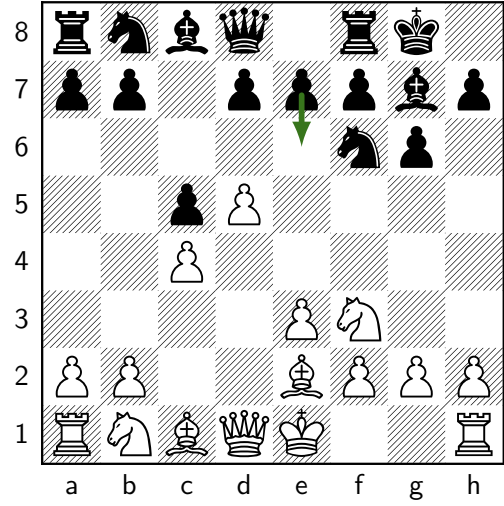


(d)

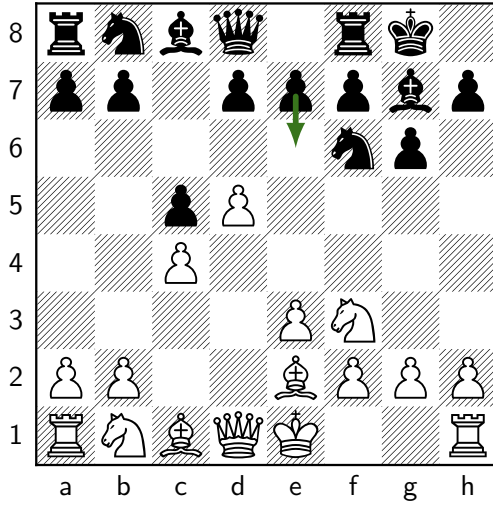
Figure 4: **Move 10, White: Kg1.** (a) Our position; Melanie plays Kg1 (**g2g1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top.p0003#989` (ply 8) [structural distance $d=59$, QAP obj=27]: the master played 0-0; under μ our *Kg2* plays the role of that game’s *Ke1*; cross-type roles: our *be7* as their *Ra1*, our *Pf2* as their *Bc1*; transferred move **g2g1 (endorses this move)**. (c) Retrieved from `ccr1_top.p0003#990` (ply 8) [structural distance $d=59$, QAP obj=27]: the master played 0-0; under μ our *Kg2* plays the role of that game’s *Ke1*; cross-type roles: our *be7* as their *Ra1*, our *Pf2* as their *Bc1*; transferred move **g2g1 (endorses this move)**. (d) Retrieved from `ccr1_top.p0000#896` (ply 11) [structural distance $d=64$, QAP obj=25]: the master played 0-0; under μ our *Kg2* plays the role of that game’s *Ke1*; cross-type roles: our *pg7* as their *Ra8*, our *Ra1* as their *Nb8*; transferred move **g2g1 (endorses this move)**. Synthesis: 4 retrieved analogs transferred to Kg1 (summed QAP objective 105, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



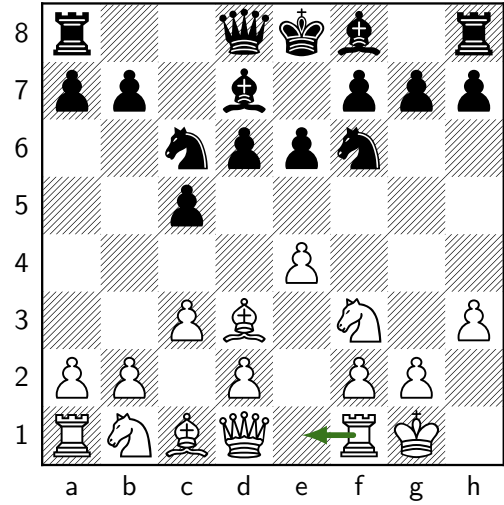
(a)



(b)

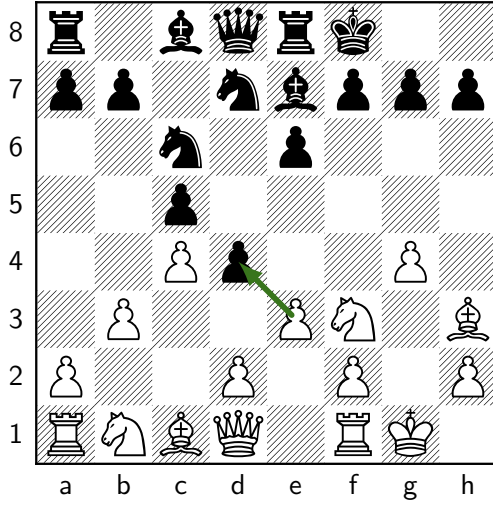


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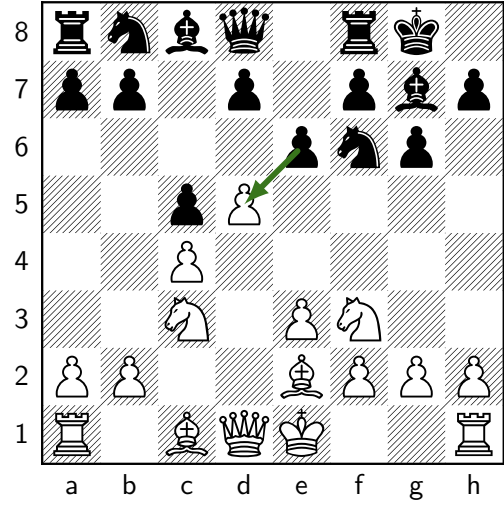


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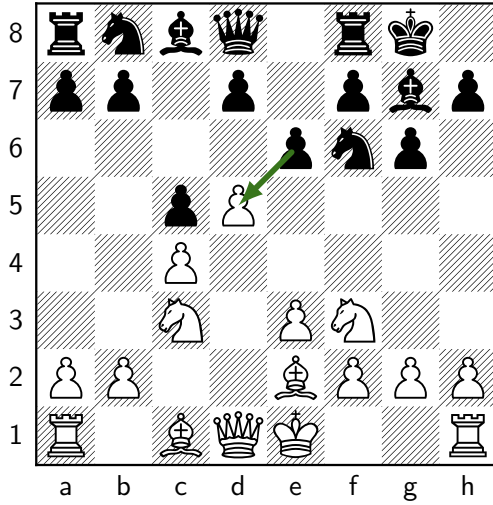
Figure 5: **Move 11, White: e3**. (a) Our position; Melanie plays **e3** (**e2e3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1281` (ply 11) [structural distance $d=56$, QAP obj=36]: the master played **e6**; under μ our *Pe2* plays the role of that game’s *Pe2*; cross-type roles: our *Qd1* as their *Ra8*, our *Pa2* as their *Nb8*; transferred move **e2e3** (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1282` (ply 11) [structural distance $d=56$, QAP obj=36]: the master played **e6**; under μ our *Pe2* plays the role of that game’s *Pe2*; cross-type roles: our *Qd1* as their *Ra8*, our *Pa2* as their *Nb8*; transferred move **e2e3** (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#1996` (ply 12) [structural distance $d=63$, QAP obj=26]: the master played **Re1**; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *Nb1* as their *Ra1*, our *pe6* as their *Nb1*; transferred move **f1e1**. Synthesis: 2 retrieved analogs transferred to **e3** (summed QAP objective 71, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



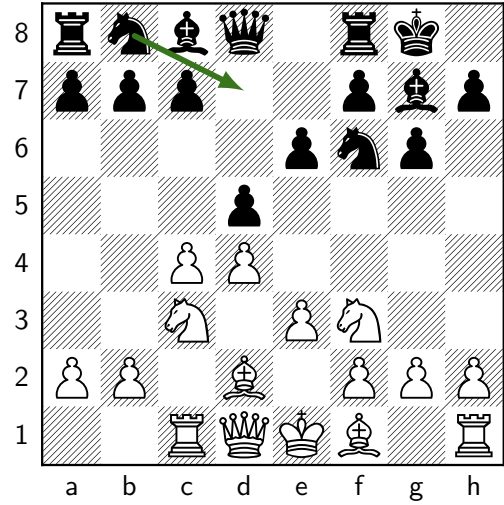
(a)



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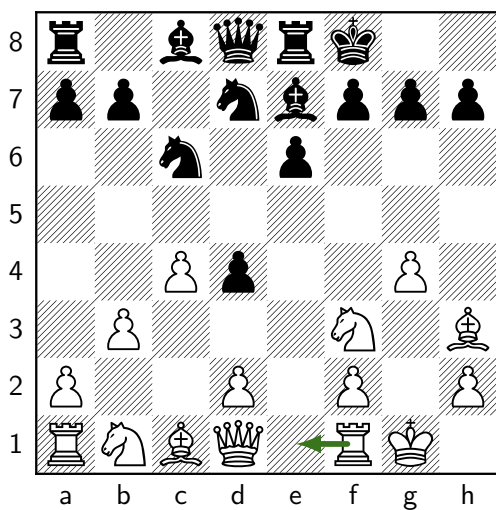


(c)

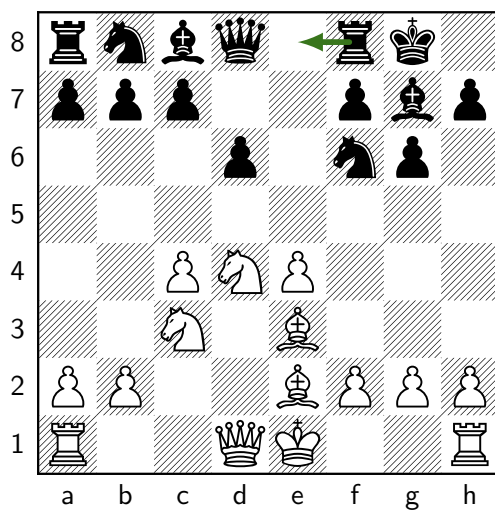


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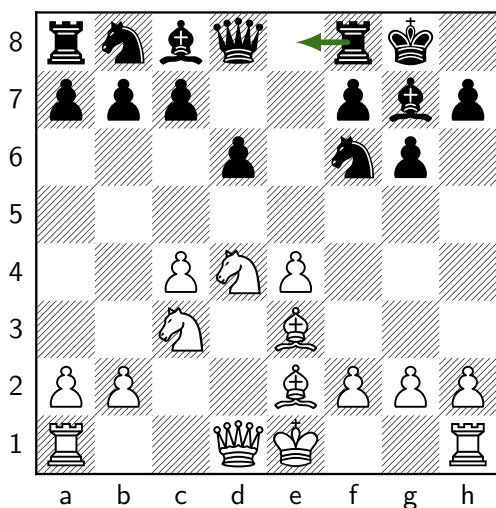
Figure 6: **Move 12, White: exd4.** (a) Our position; Melanie plays **exd4** (**e3d4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1281` (ply 13) [structural distance $d=41$, QAP obj=92]: the master played **exd5**; under μ our *Pe3* plays the role of that game’s *Pe3*; cross-type roles: our *pb7* as their *Ra8*, our *Pa2* as their *Qd8*; transferred move **e3d4** (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1282` (ply 13) [structural distance $d=41$, QAP obj=92]: the master played **exd5**; under μ our *Pe3* plays the role of that game’s *Pe3*; cross-type roles: our *pb7* as their *Ra8*, our *Pa2* as their *Qd8*; transferred move **e3d4** (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1421` (ply 13) [structural distance $d=47$, QAP obj=86]: the master played **Nbd7**; under μ our *Nb1* plays the role of that game’s *Nb1*; cross-type roles: our *Bh3* as their *Ra8*, our *ra8* as their *Qd8*; transferred move **b1a3**. Synthesis: 2 retrieved analogs transferred to **exd4** (summed QAP objective 185, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



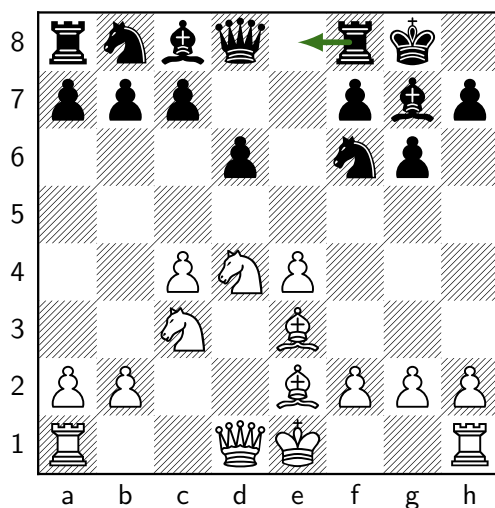
(a)



(b)

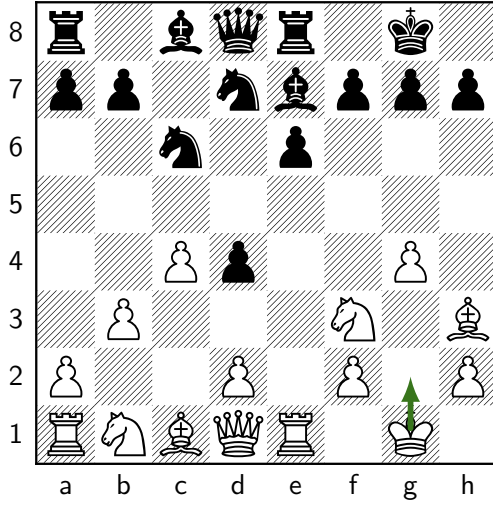


(c)

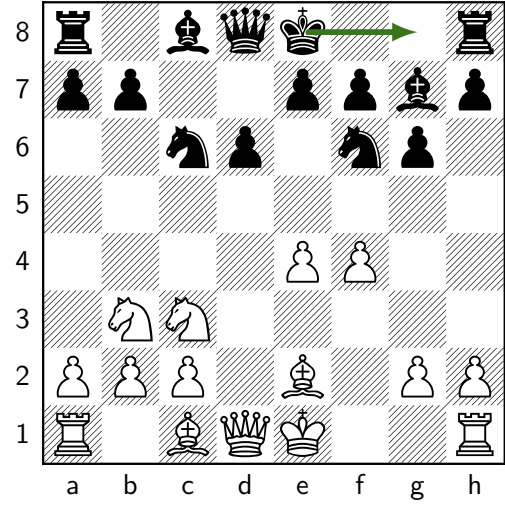


(d)

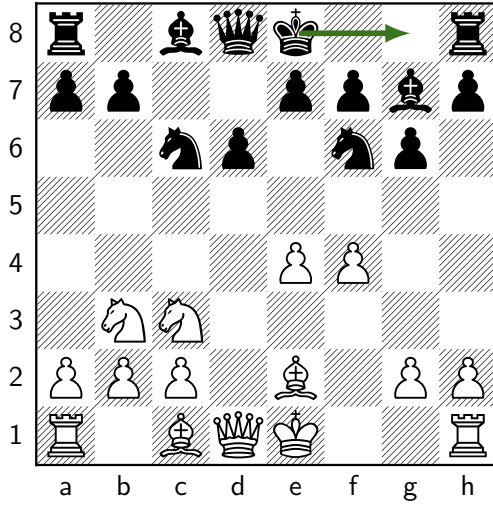
Figure 7: **Move 13, White: Re1.** (a) Our position; Melanie plays **Re1** (**f1e1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3999` (ply 15) [structural distance $d=52$, QAP obj=27]: the master played **Re8**; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *Nb1* as their *Ra8*, our *Bc1* as their *Nb8*; transferred move **f1e1** (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#4000` (ply 15) [structural distance $d=52$, QAP obj=27]: the master played **Re8**; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *Nb1* as their *Ra8*, our *Bc1* as their *Nb8*; transferred move **f1e1** (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1624` (ply 15) [structural distance $d=52$, QAP obj=27]: the master played **Re8**; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *Nb1* as their *Ra8*, our *Bc1* as their *Nb8*; transferred move **f1e1** (**endorses this move**). Synthesis: 4 retrieved analogs transferred to **Re1** (summed QAP objective 107, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



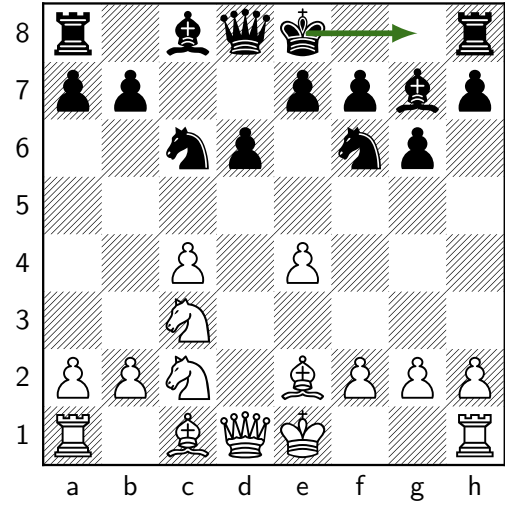
(a)



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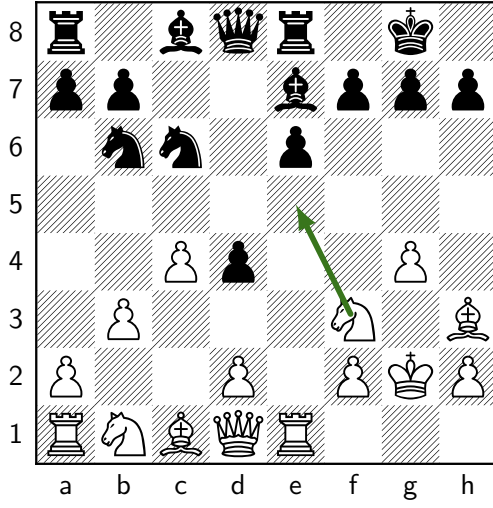


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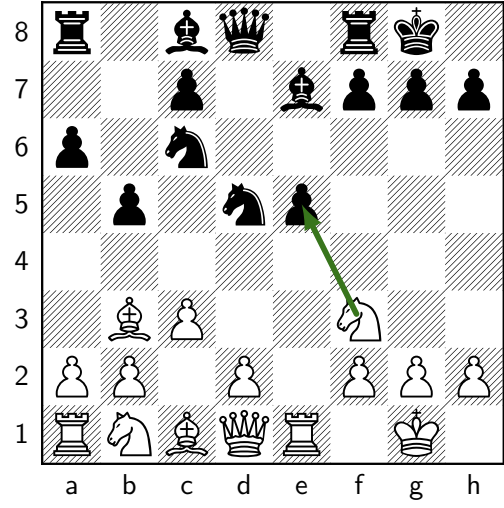


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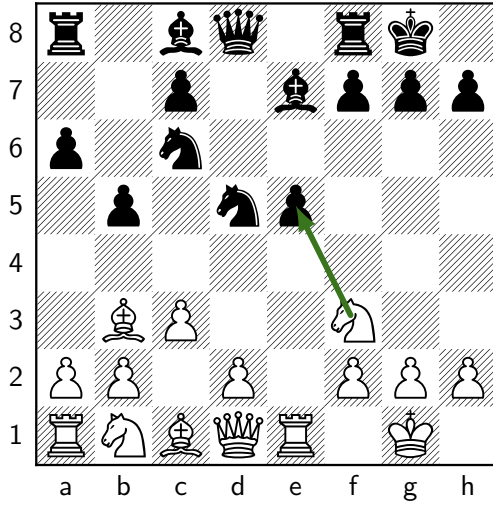
Figure 8: **Move 14, White: Kg2.** (a) Our position; Melanie plays Kg2 (**g1g2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1090` (ply 15) [structural distance $d=72$, QAP obj=53]: the master played 0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Bc1* as their *Pa7*, our *Qd1* as their *Pb7*; transferred move **g1g2 (endorses this move)**. (c) Retrieved from `ccrl_top_p0000#1091` (ply 15) [structural distance $d=72$, QAP obj=53]: the master played 0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Bc1* as their *Pa7*, our *Qd1* as their *Pb7*; transferred move **g1g2 (endorses this move)**. (d) Retrieved from `ccrl_top_p0000#708` (ply 15) [structural distance $d=68$, QAP obj=53]: the master played 0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Bc1* as their *Pa7*, our *Qd1* as their *Pb7*; transferred move **g1g2 (endorses this move)**. Synthesis: 4 retrieved analogs transferred to Kg2 (summed QAP objective 212, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



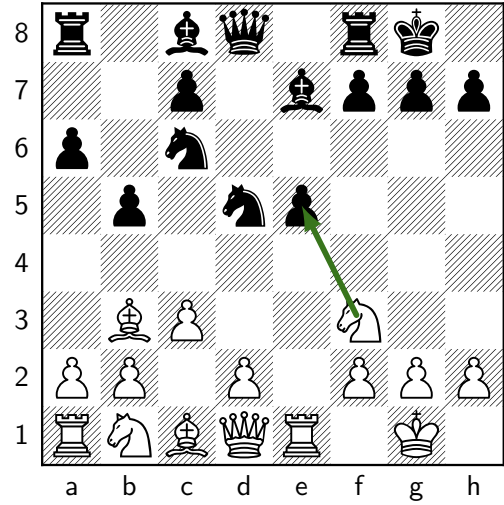
(a)



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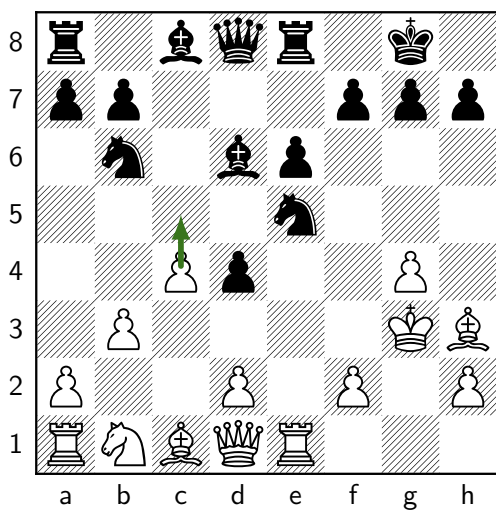


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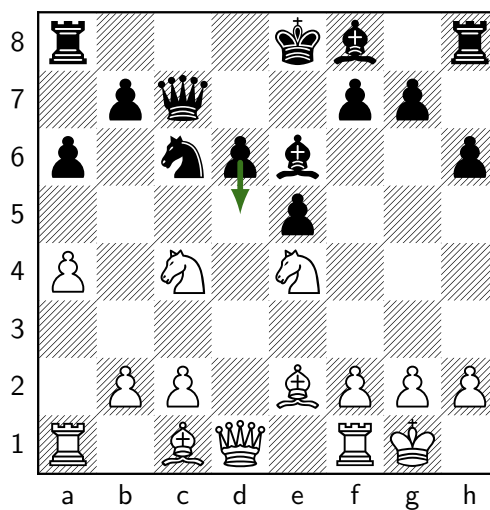


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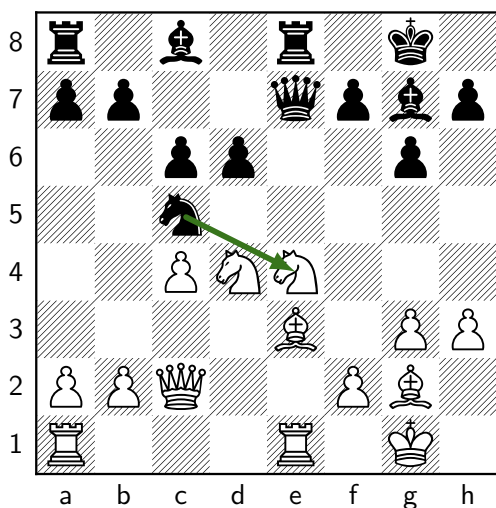
Figure 9: **Move 15, White: Ne5.** (a) Our position; Melanie plays Ne5 (f3e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#414` (ply 18) [structural distance $d=60$, QAP obj=30]: the master played Nxe5; under μ our Nf3 plays the role of that game’s Nf3; cross-type roles: our Nb1 as their Ra1, our Pg4 as their Nb1; transferred move f3e5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#415` (ply 18) [structural distance $d=60$, QAP obj=30]: the master played Nxe5; under μ our Nf3 plays the role of that game’s Nf3; cross-type roles: our Nb1 as their Ra1, our Pg4 as their Nb1; transferred move f3e5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#112` (ply 18) [structural distance $d=60$, QAP obj=30]: the master played Nxe5; under μ our Nf3 plays the role of that game’s Nf3; cross-type roles: our Nb1 as their Ra1, our Pg4 as their Nb1; transferred move f3e5 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Ne5 (summed QAP objective 266, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



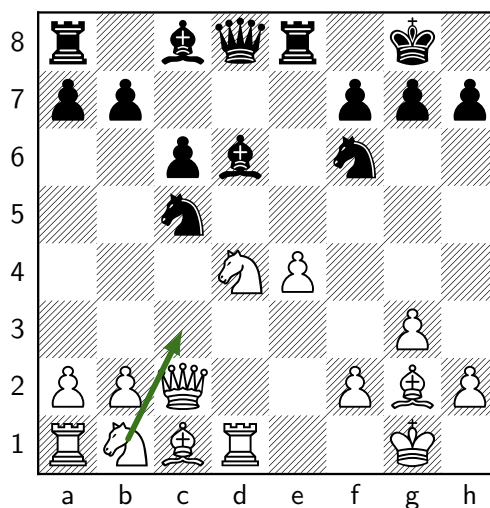
(a)



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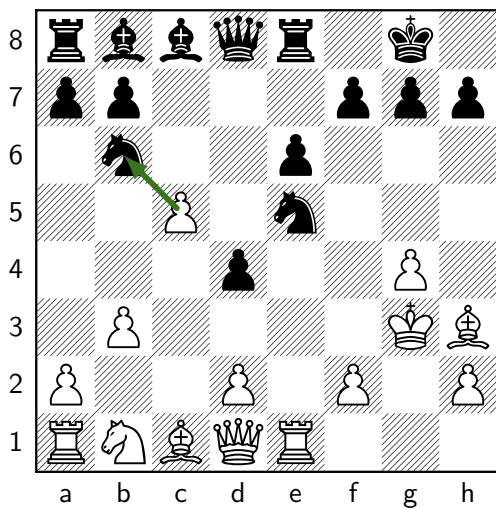


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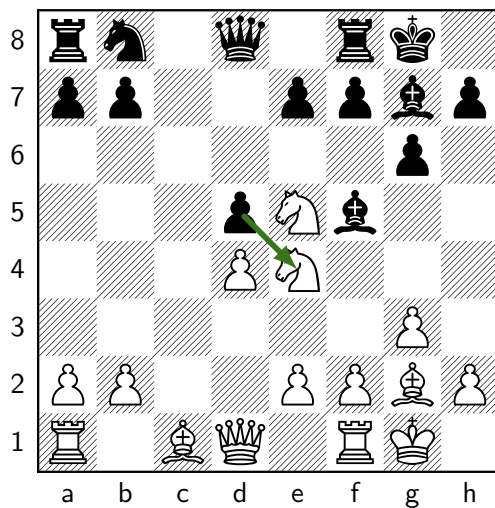


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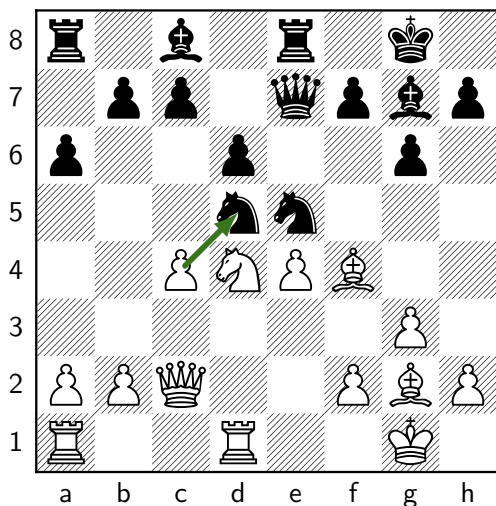
Figure 10: **Move 17, White: c5.** (a) Our position; Melanie plays c5 (c4c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2458` (ply 23) [structural distance $d=78$, QAP obj=46]: the master played d5; under μ our *Pc4* plays the role of that game’s *Pd3*; cross-type roles: our *Pf2* as their *Ra8*, our *Ph2* as their *Rh8*; transferred move c4c5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2043` (ply 27) [structural distance $d=75$, QAP obj=31]: the master played Nxe4; under μ our *Pc4* plays the role of that game’s *Nc4* (a **cross-type** correspondence); cross-type roles: our *Qd1* as their *Re8*, our *Re1* as their *Qe7*; transferred move c4c5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2823` (ply 24) [structural distance $d=78$, QAP obj=53]: the master played Nc3; under μ our *Qd1* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *Pf2* as their *Ra1*, our *Ph2* as their *Rd1*; transferred move d1e2. Synthesis: 2 retrieved analogs transferred to c5 (summed QAP objective 78, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



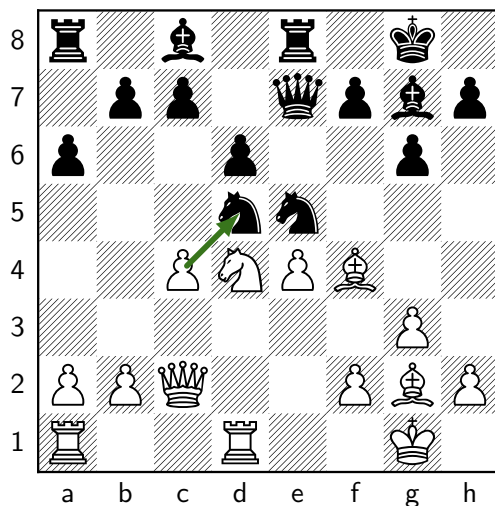
(a)



(b)

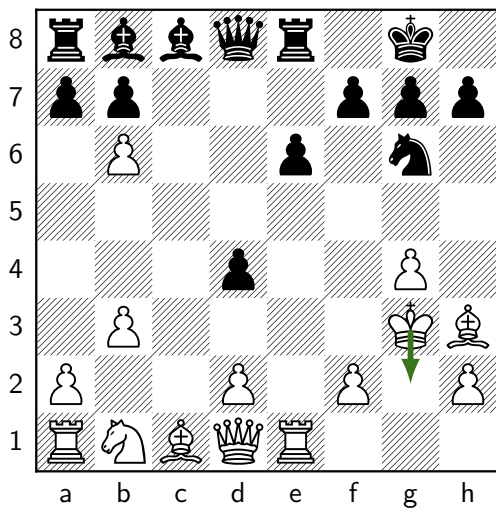


(c)

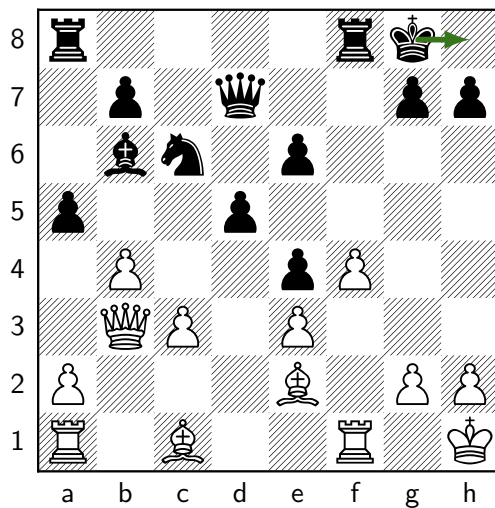


(d)

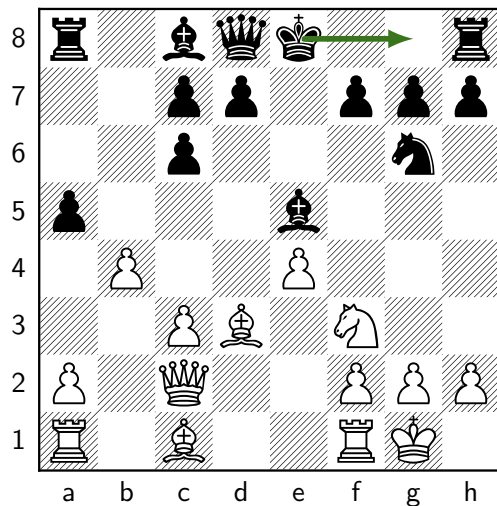
Figure 11: **Move 18, White: cxb6.** (a) Our position; Melanie plays cxb6 (c5b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#2563 (ply 19) [structural distance $d=87$, QAP obj=51]: the master played dxe4; under μ our $Pc5$ plays the role of that game’s $Pd4$; cross-type roles: our $Nb1$ as their $Ra8$, our $Pd2$ as their $Nb8$; transferred move c5b6 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#1296 (ply 26) [structural distance $d=85$, QAP obj=47]: the master played cxd5; under μ our $Pc5$ plays the role of that game’s $Pc4$; cross-type roles: our $pd4$ as their $Ra1$, our $Ph2$ as their $Rd1$; transferred move c5b6 (**endorses this move**). (d) Retrieved from ccr1_top_p0003#2214 (ply 26) [structural distance $d=85$, QAP obj=47]: the master played cxd5; under μ our $Pc5$ plays the role of that game’s $Pc4$; cross-type roles: our $pd4$ as their $Ra1$, our $Ph2$ as their $Rd1$; transferred move c5b6 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to cxb6 (summed QAP objective 192, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



(a)

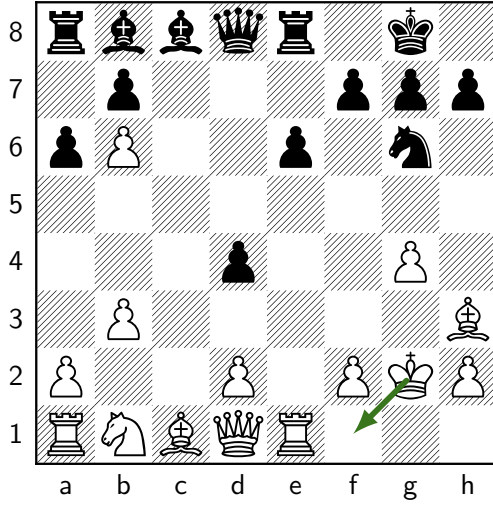


(b)

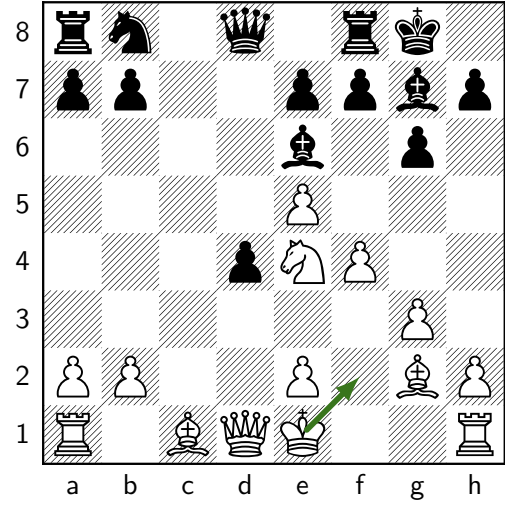


(c)

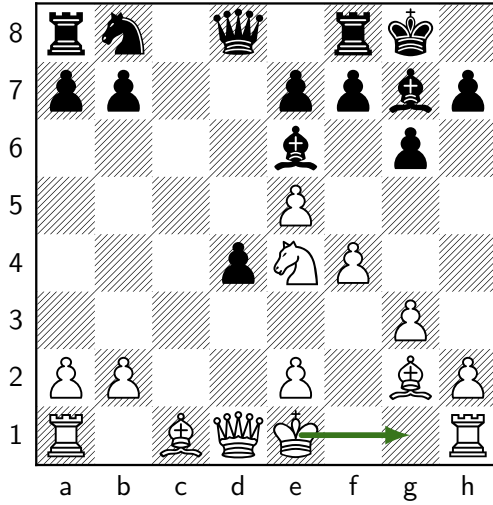
Figure 12: **Move 19, White: Kg2.** (a) Our position; Melanie plays Kg2 ($g3g2$). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1357` (ply 31) [structural distance $d=79$, QAP obj=31]: the master played $Kh8$; under μ our $Kg3$ plays the role of that game’s $Kg1$; cross-type roles: our $Qd1$ as their $Ra8$, our $Bh3$ as their $Qd7$; transferred move $g3g2$ (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3858` (ply 23) [structural distance $d=84$, QAP obj=16]: the master played $0-0$; under μ our $Kg3$ plays the role of that game’s $Ke1$; cross-type roles: our $Pg4$ as their $Ra8$, our $Ph2$ as their $Qd8$; transferred move $g3g2$ (**endorses this move**). Synthesis: 2 retrieved analogs transferred to Kg2 (summed QAP objective 47, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



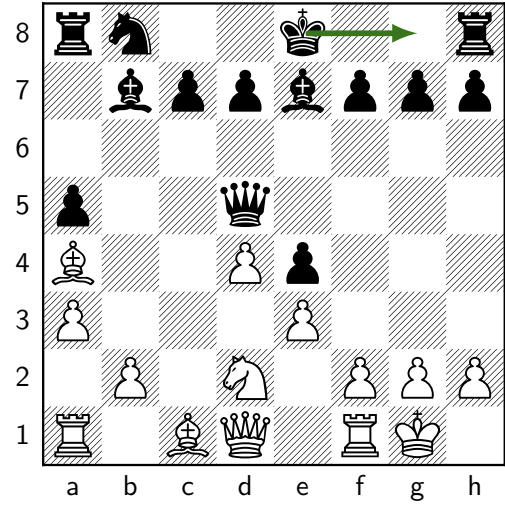
(a)



(b)

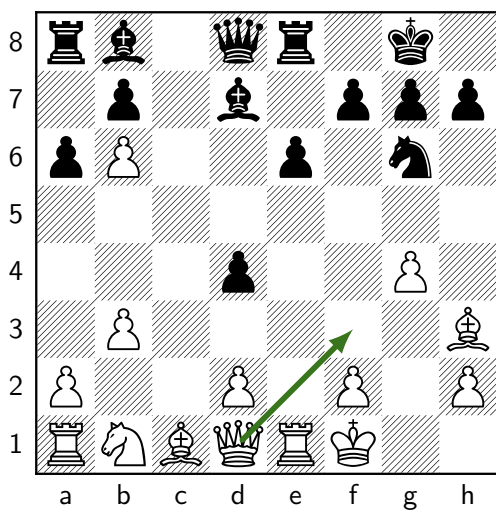


(c)

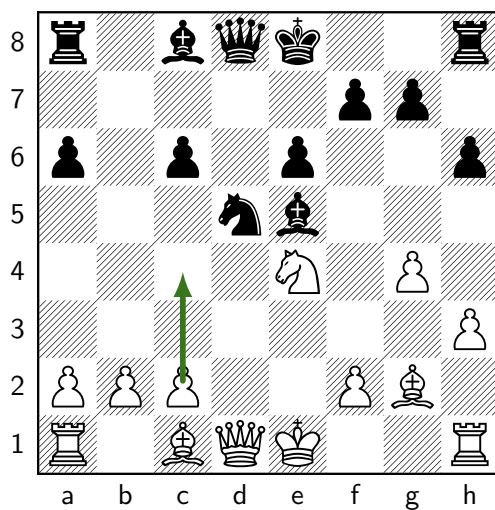


(d)

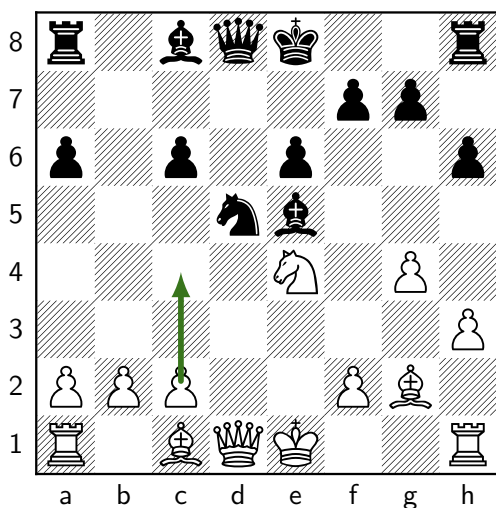
Figure 13: **Move 20, White: Kf1**. (a) Our position; Melanie plays Kf1 (g2f1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2853` (ply 22) [structural distance $d=57$, QAP obj=43]: the master played Kf2; under μ our Kg2 plays the role of that game’s Ke1; cross-type roles: our Qd1 as their Bc1, our Re1 as their Qd1; transferred move g2f1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2430` (ply 22) [structural distance $d=57$, QAP obj=43]: the master played 0-0; under μ our Kg2 plays the role of that game’s Ke1; cross-type roles: our Qd1 as their Bc1, our Re1 as their Qd1; transferred move g2f1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2926` (ply 29) [structural distance $d=74$, QAP obj=15]: the master played 0-0; under μ our Kg2 plays the role of that game’s Ke1; cross-type roles: our bc8 as their Ra8, our Ra1 as their Nb8; transferred move g2f1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kf1 (summed QAP objective 101, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



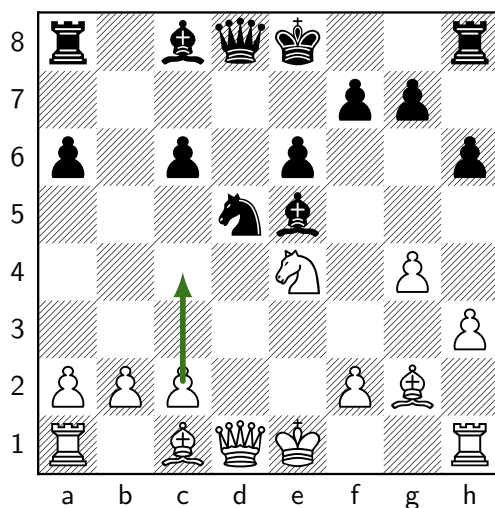
(a)



(b)

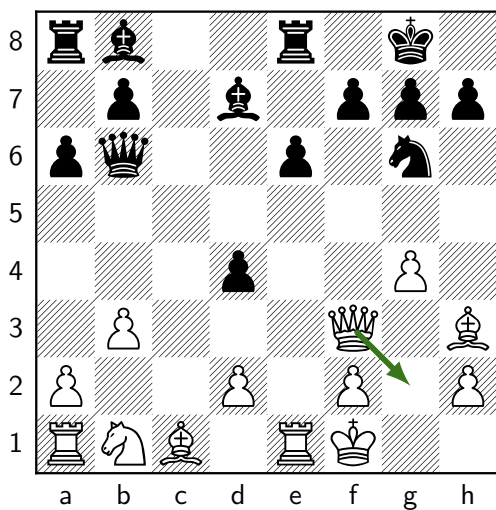


(c)

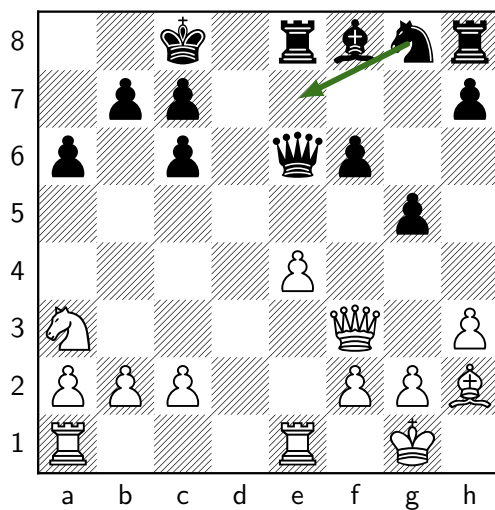


(d)

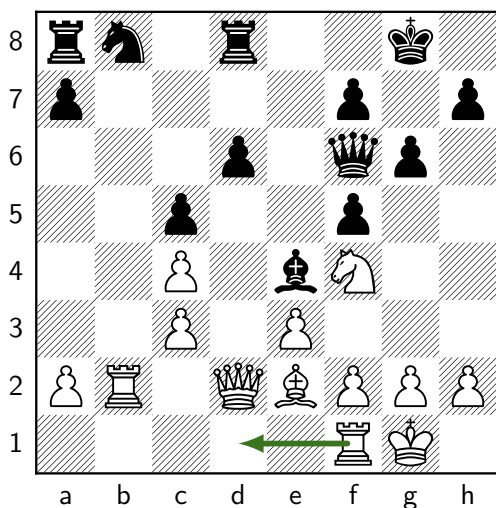
Figure 14: **Move 21, White: Qf3**. (a) Our position; Melanie plays Qf3 (d1f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2025` (ply 24) [structural distance $d=61$, QAP obj=33]: the master played c4; under μ our Qd1 plays the role of that game’s Pc2 (a **cross-type** correspondence); cross-type roles: our Pb3 as their Ra1, our Re1 as their Qd1; transferred move d1f3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2026` (ply 24) [structural distance $d=61$, QAP obj=33]: the master played c4; under μ our Qd1 plays the role of that game’s Pc2 (a **cross-type** correspondence); cross-type roles: our Pb3 as their Ra1, our Re1 as their Qd1; transferred move d1f3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3523` (ply 24) [structural distance $d=61$, QAP obj=33]: the master played c4; under μ our Qd1 plays the role of that game’s Pc2 (a **cross-type** correspondence); cross-type roles: our Pb3 as their Ra1, our Re1 as their Qd1; transferred move d1f3 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qf3 (summed QAP objective 130, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



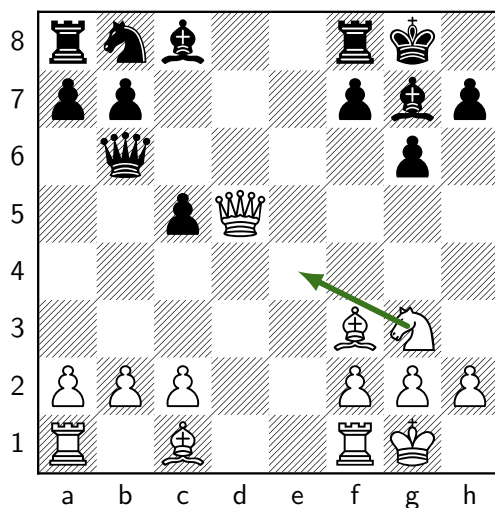
(a)



(b)

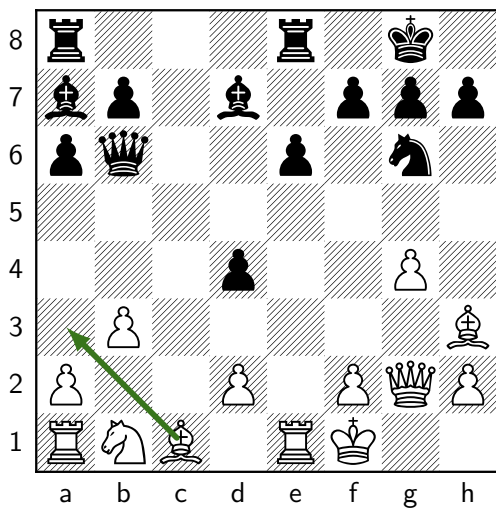


(c)

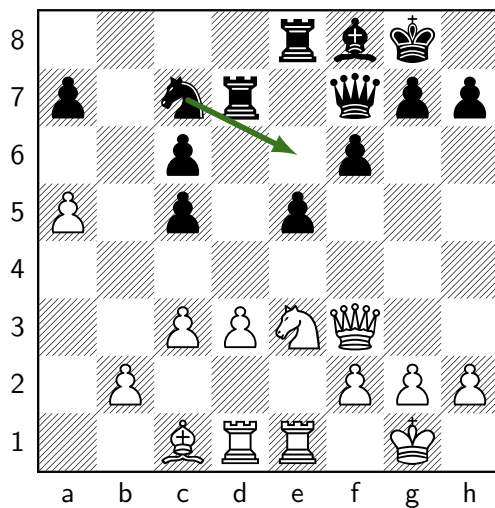


(d)

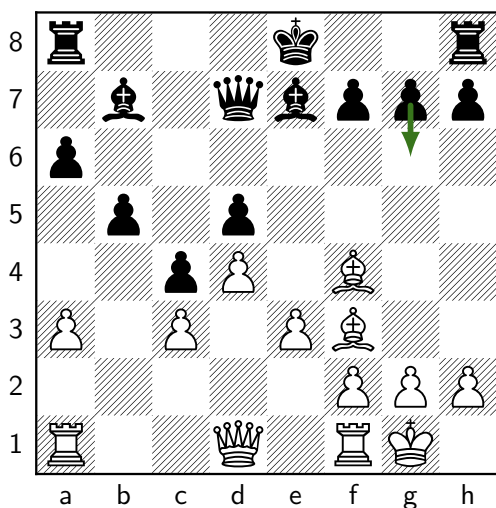
Figure 15: **Move 22, White: Qg2.** (a) Our position; Melanie plays Qg2 (f3g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1500` (ply 31) [structural distance $d=86$, QAP obj=88]: the master played Ne7; under μ our Pg4 plays the role of that game's Ng1 (a **cross-type** correspondence); cross-type roles: our Bh3 as their Re8, our Re1 as their Bf8; transferred move g4g5. (c) Retrieved from `ccr1_top_p0003#2477` (ply 30) [structural distance $d=74$, QAP obj=88]: the master played Rd1; under μ our Re1 plays the role of that game's Rf1; cross-type roles: our Nb1 as their Pa2, our bb8 as their Rb2; transferred move e1d1. (d) Retrieved from `ccr1_top_p0001#2499` (ply 26) [structural distance $d=75$, QAP obj=73]: the master played Ne4; under μ our Nb1 plays the role of that game's Ng3; cross-type roles: our Pa2 as their Ra1, our ng6 as their Bc1; transferred move b1c3. Synthesis: no retrieved analog transferred to Qg2 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



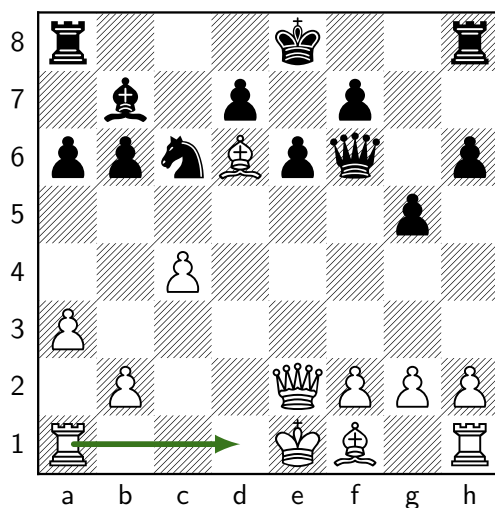
(a)



(b)

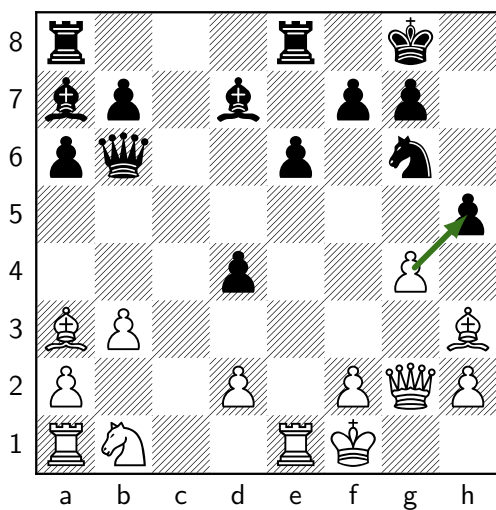


(c)

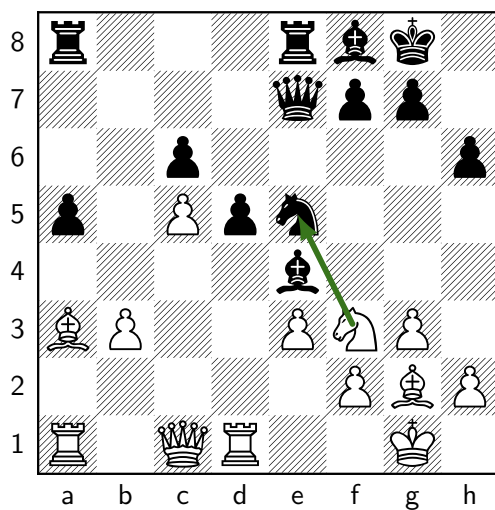


(d)

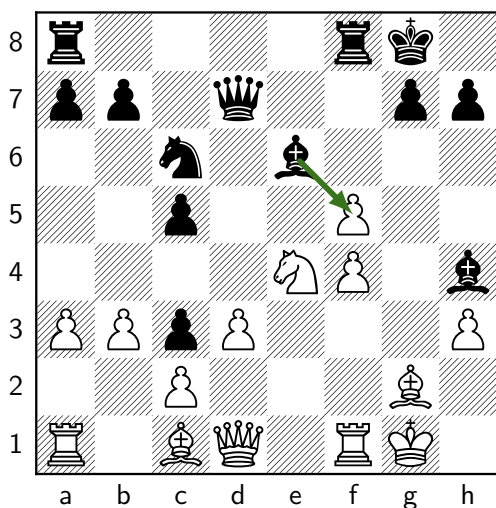
Figure 16: **Move 23, White: Ba3.** (a) Our position; Melanie plays Ba3 (c1a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#2072` (ply 45) [structural distance $d=82$, QAP obj=92]: the master played Ne6; under μ our Bc1 plays the role of that game’s Nc2 (a **cross-type** correspondence); cross-type roles: our Bh3 as their Re8, our Qg2 as their Bf8; transferred move c1a3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#47` (ply 29) [structural distance $d=79$, QAP obj=44]: the master played g6; under μ our Bc1 plays the role of that game’s Pg2 (a **cross-type** correspondence); cross-type roles: our Pd2 as their Ra8, our Qg2 as their Rh8; transferred move c1a3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#10` (ply 30) [structural distance $d=63$, QAP obj=104]: the master played Rd1; under μ our Re1 plays the role of that game’s Ra1; cross-type roles: our Pg4 as their Rh1, our Ra1 as their Ph2; transferred move e1e2. Synthesis: 2 retrieved analogs transferred to Ba3 (summed QAP objective 137, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



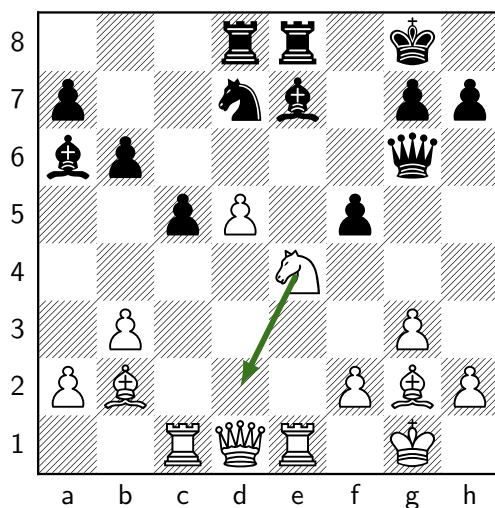
(a)



(b)

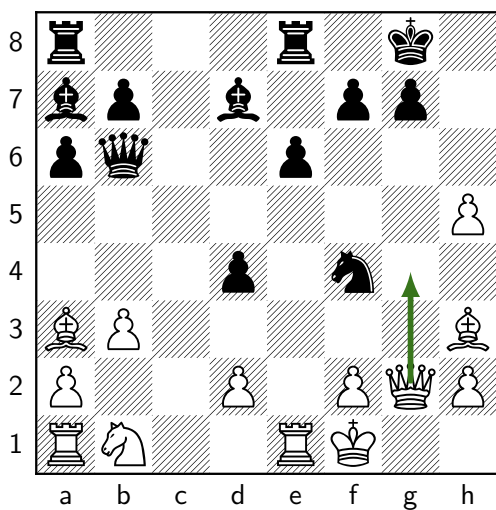


(c)

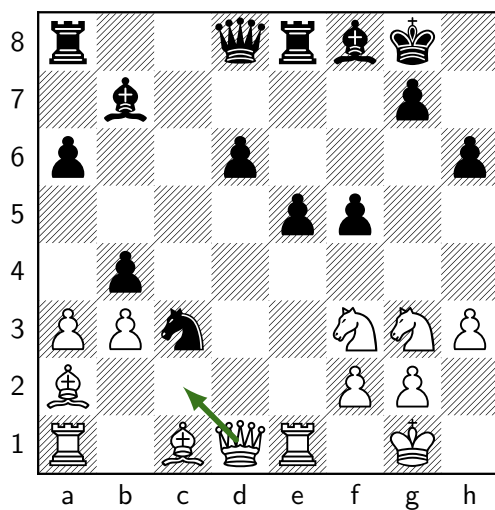


(d)

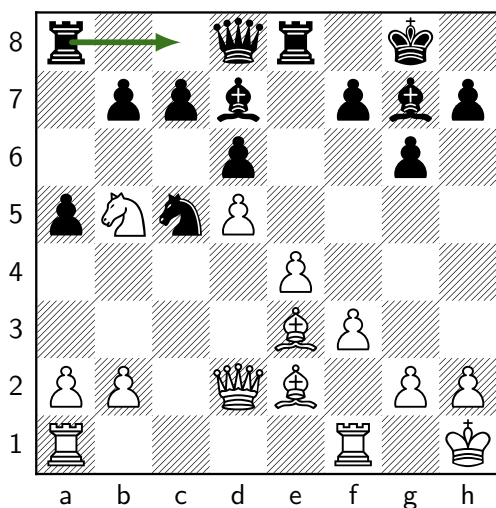
Figure 17: **Move 24, White: gxh5.** (a) Our position; Melanie plays gxh5 (g4h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#663 (ply 54) [structural distance $d=73$, QAP obj=38]: the master played Nxe5; under μ our $Pg4$ plays the role of that game’s $Nf3$ (a **cross-type** correspondence); cross-type roles: our $Bh3$ as their $Qc1$, our $Qg2$ as their $Bg2$; transferred move g4h5 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#2259 (ply 33) [structural distance $d=88$, QAP obj=37]: the master played Bxf5; under μ our $Pg4$ plays the role of that game’s $Be3$ (a **cross-type** correspondence); cross-type roles: our $Pa2$ as their $Ra8$, our $Qg2$ as their $Rf8$; transferred move g4h5 (**endorses this move**). (d) Retrieved from ccr1_top_p0003#1975 (ply 36) [structural distance $d=82$, QAP obj=34]: the master played Nd2; under μ our $Pg4$ plays the role of that game’s $Ne4$ (a **cross-type** correspondence); cross-type roles: our $Ba3$ as their $Re1$, our $Nb1$ as their $Pa2$; transferred move g4h5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to gxh5 (summed QAP objective 109, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



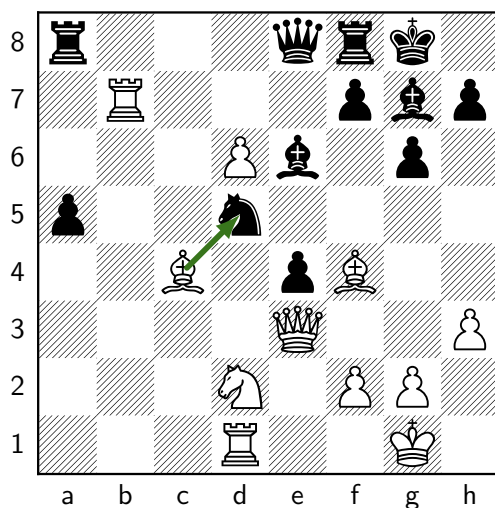
(a)



(b)

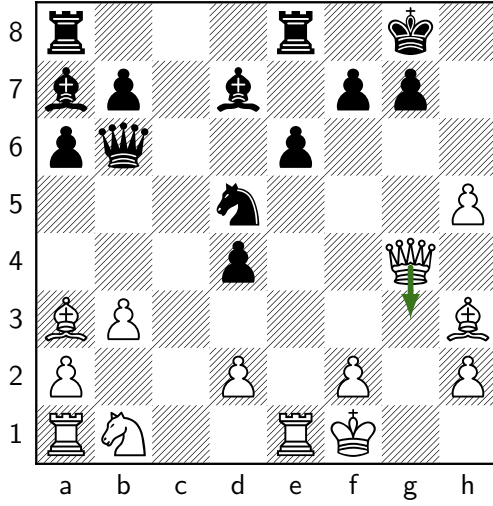


(c)

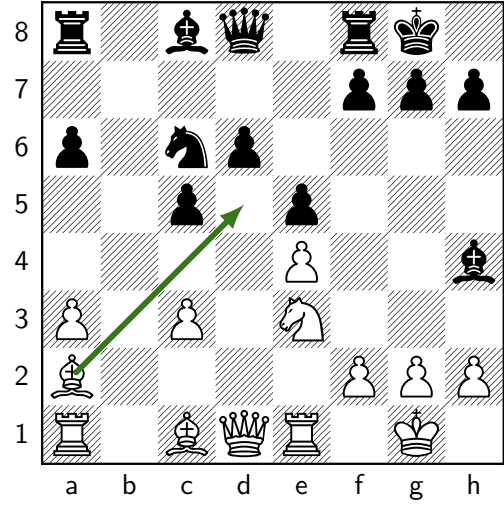


(d)

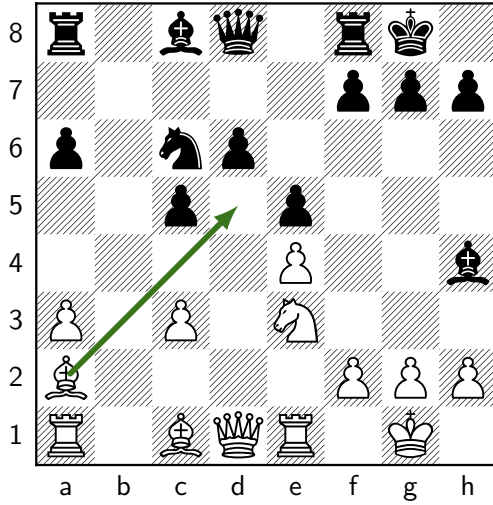
Figure 18: **Move 25, White: Qg4.** (a) Our position; Melanie plays Qg4 (g2g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#919` (ply 40) [structural distance $d=102$, QAP obj=86]: the master played Qc2; under μ our Qg2 plays the role of that game’s Qd1; cross-type roles: our Ph2 as their Ra1, our Pa2 as their Bc1; transferred move g2h1. (c) Retrieved from `ccr1_top_p0000#1121` (ply 33) [structural distance $d=97$, QAP obj=56]: the master played Rc8; under μ our Ba3 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our ba7 as their Qd8, our Ra1 as their Pb7; transferred move a3c1. (d) Retrieved from `ccr1_top_p0001#2790` (ply 56) [structural distance $d=106$, QAP obj=56]: the master played Bxd5; under μ our Nb1 plays the role of that game’s Bc4 (a **cross-type** correspondence); cross-type roles: our pd4 as their Nd2, our Ba3 as their pa5; transferred move b1c3. Synthesis: no retrieved analog transferred to Qg4 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



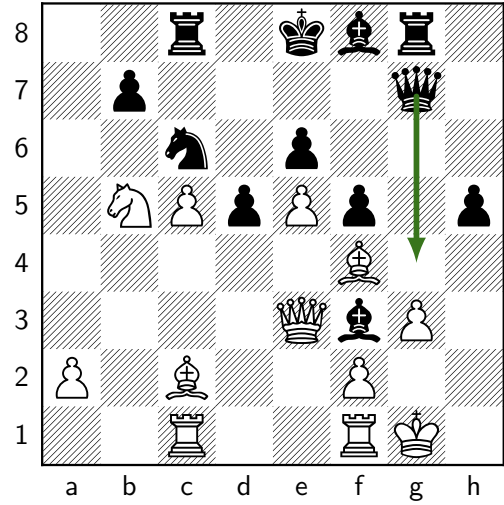
(a)



(b)

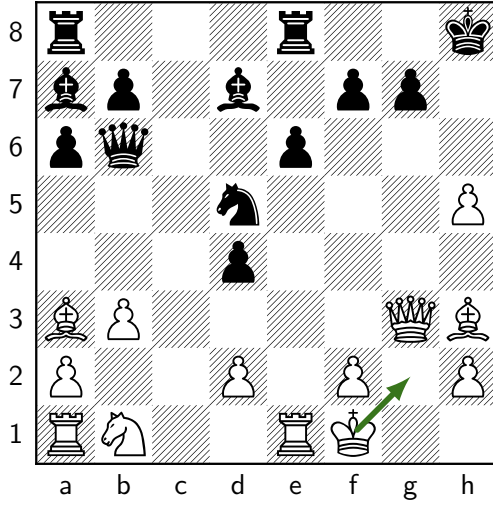


(c)

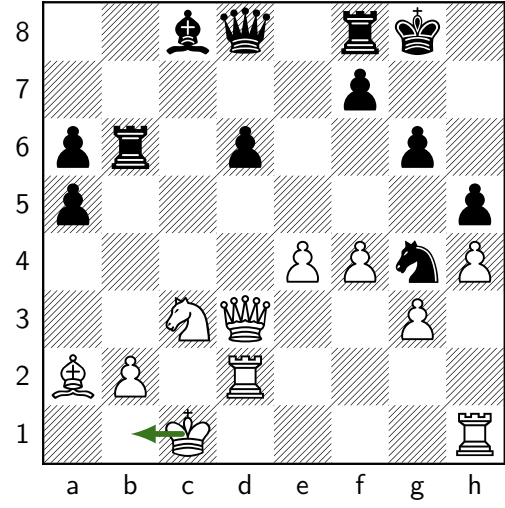


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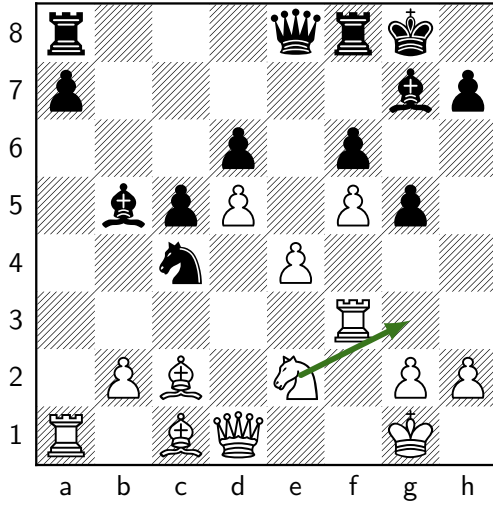
Figure 19: **Move 26, White: Qg3.** (a) Our position; Melanie plays Qg3 (g4g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#824` (ply 34) [structural distance $d=77$, QAP obj=75]: the master played Bd5; under μ our Qg4 plays the role of that game’s Ba2 (a **cross-type** correspondence); cross-type roles: our Ba3 as their Ra1, our Re1 as their Bc1; transferred move g4g3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2083` (ply 34) [structural distance $d=77$, QAP obj=75]: the master played Bd5; under μ our Qg4 plays the role of that game’s Ba2 (a **cross-type** correspondence); cross-type roles: our Ba3 as their Ra1, our Re1 as their Bc1; transferred move g4g3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#882` (ply 51) [structural distance $d=80$, QAP obj=33]: the master played Qg4; under μ our Qg4 plays the role of that game’s Qg2; cross-type roles: our Bh3 as their Rc8, our Nb1 as their Bf8; transferred move g4g3 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Qg3 (summed QAP objective 183, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



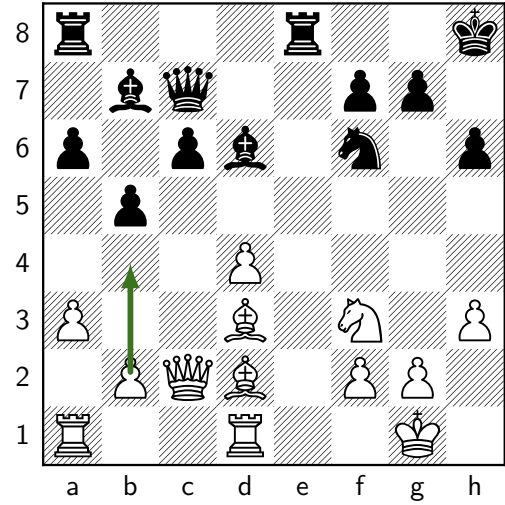
(a)



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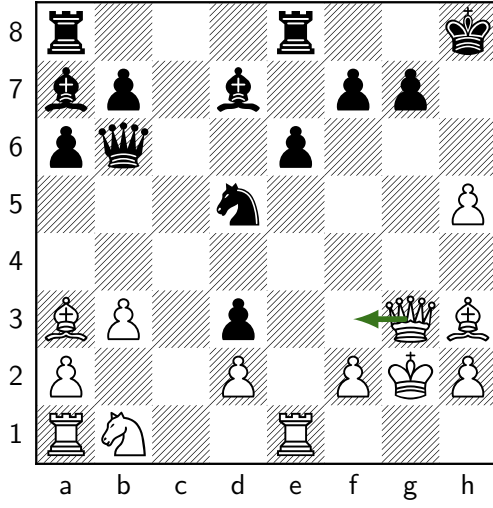


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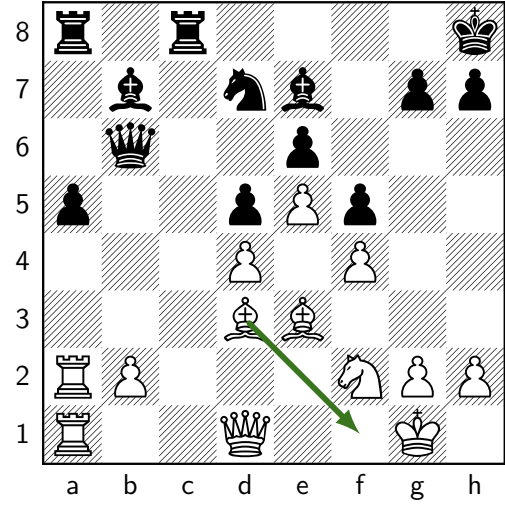


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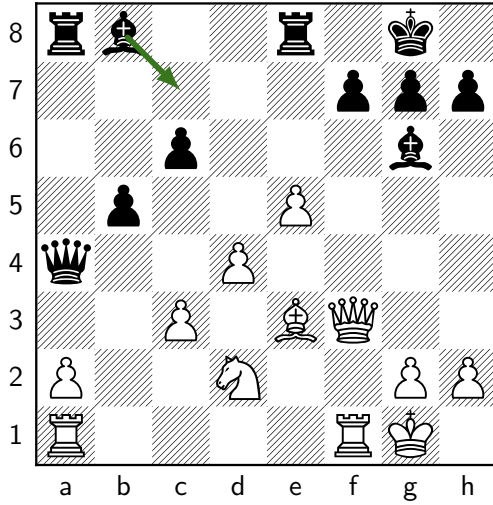
Figure 20: **Move 27, White: Kg2.** (a) Our position; Melanie plays Kg2 (f1g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#563` (ply 52) [structural distance $d=92$, QAP obj=112]: the master played Kb1; under μ our Kf1 plays the role of that game’s Kc1; cross-type roles: our pf7 as their Rh1, our Re1 as their Ba2; transferred move f1g2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2779` (ply 44) [structural distance $d=94$, QAP obj=65]: the master played Ng3; under μ our Ph5 plays the role of that game’s Ne2 (a **cross-type** correspondence); cross-type roles: our Qg3 as their Bc1, our ra8 as their Qd1; transferred move h5h6. (d) Retrieved from `ccr1_top_p0002#2598` (ply 38) [structural distance $d=82$, QAP obj=55]: the master played b4; under μ our Nb1 plays the role of that game’s Pb2 (a **cross-type** correspondence); cross-type roles: our Pd2 as their Ra1, our Pf2 as their Rd1; transferred move b1c3. Synthesis: 1 retrieved analog transferred to Kg2 (summed QAP objective 112, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



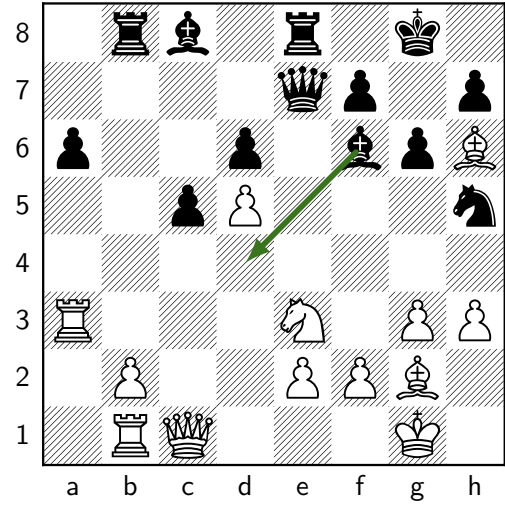
(a)



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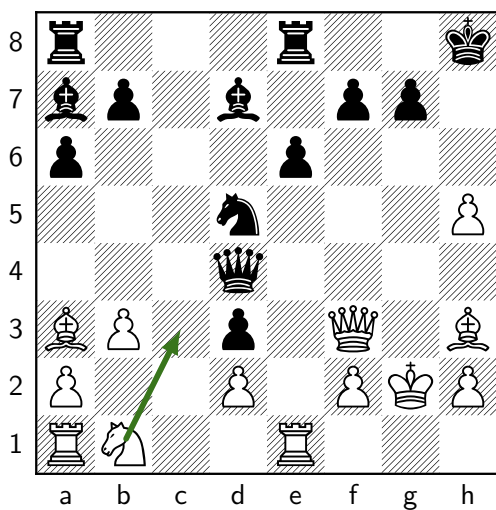


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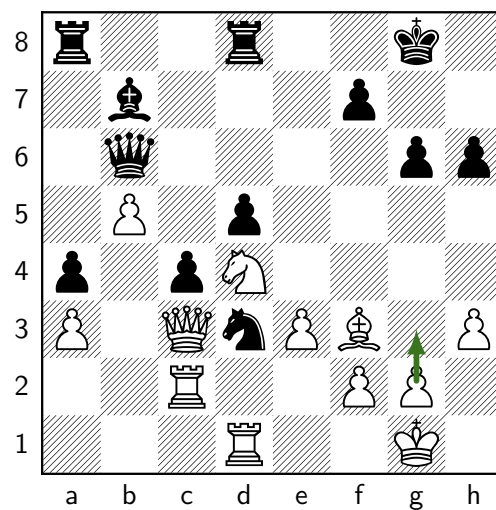


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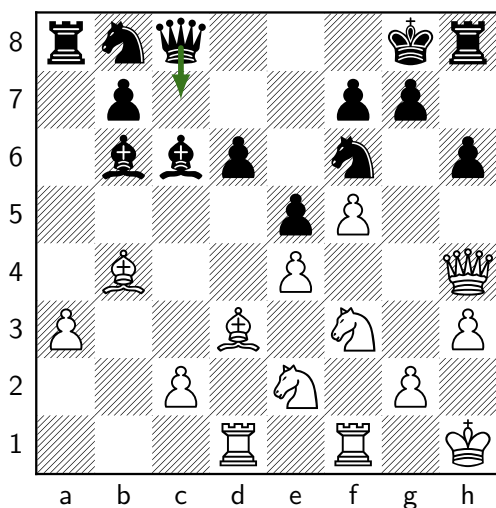
Figure 21: **Move 28, White: Qf3**. (a) Our position; Melanie plays Qf3 (g3f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2358` (ply 52) [structural distance $d=109$, QAP obj=97]: the master played Bf1; under μ our Qg3 plays the role of that game’s Bd3 (a **cross-type** correspondence); cross-type roles: our Ph2 as their Ra1, our Re1 as their Qd1; transferred move g3f3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#750` (ply 47) [structural distance $d=104$, QAP obj=122]: the master played Bc7; under μ our Bh3 plays the role of that game’s Bb1; cross-type roles: our ba7 as their Ra8, our Bh3 as their Re8; transferred move h3g4. (d) Retrieved from `ccr1_top_p0001#523` (ply 51) [structural distance $d=104$, QAP obj=63]: the master played Bd4; under μ our Bh3 plays the role of that game’s Bf3; cross-type roles: our nd5 as their Bc8, our pb7 as their Re8; transferred move h3g4. Synthesis: 1 retrieved analog transferred to Qf3 (summed QAP objective 97, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



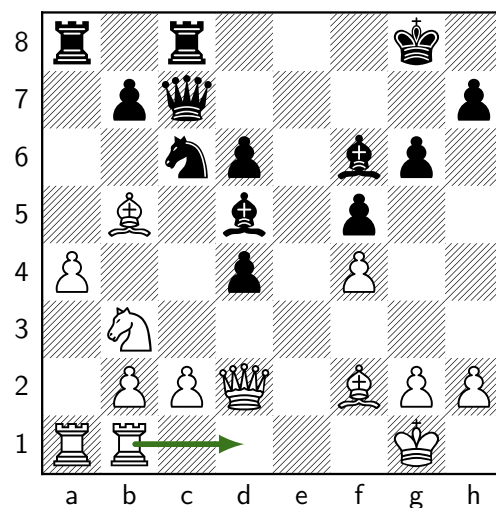
(a)



(b)

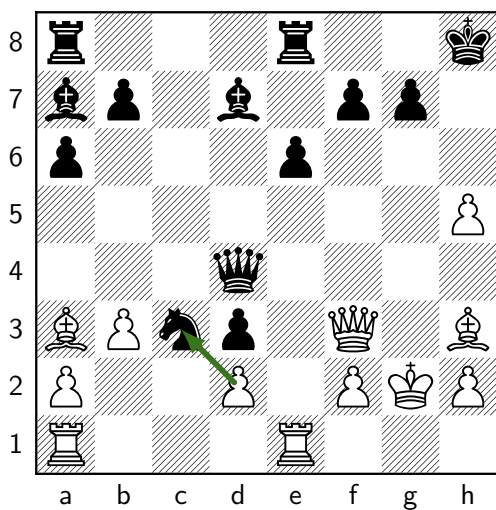


(c)

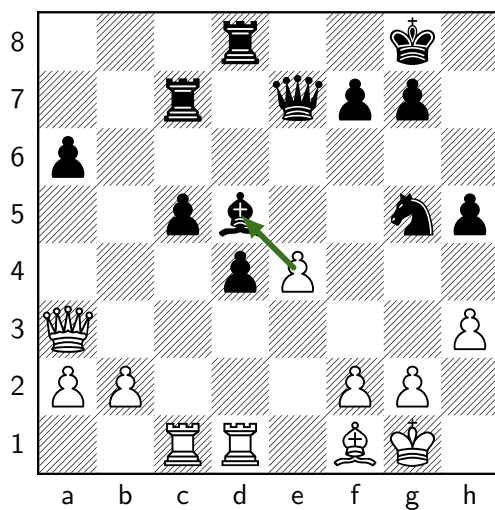


(d)

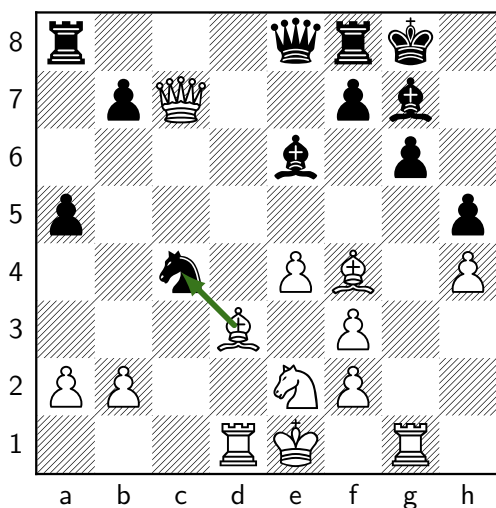
Figure 22: **Move 29, White: Nc3.** (a) Our position; Melanie plays Nc3 (b1c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2023` (ply 50) [structural distance $d=110$, QAP obj=53]: the master played g3; under μ our Nb1 plays the role of that game’s Pg2 (a **cross-type** correspondence); cross-type roles: our Qf3 as their Rd1, our Pf2 as their Rc2; transferred move b1c3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#26` (ply 43) [structural distance $d=101$, QAP obj=48]: the master played Qc7; under μ our Nb1 plays the role of that game’s Qc1 (a **cross-type** correspondence); cross-type roles: our pf7 as their Nb8, our Ra1 as their Pe5; transferred move b1c3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3145` (ply 40) [structural distance $d=112$, QAP obj=40]: the master played Rd1; under μ our Nb1 plays the role of that game’s Rb1 (a **cross-type** correspondence); cross-type roles: our bd7 as their Ra1, our Bh3 as their Qd2; transferred move b1c3 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Nc3 (summed QAP objective 141, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



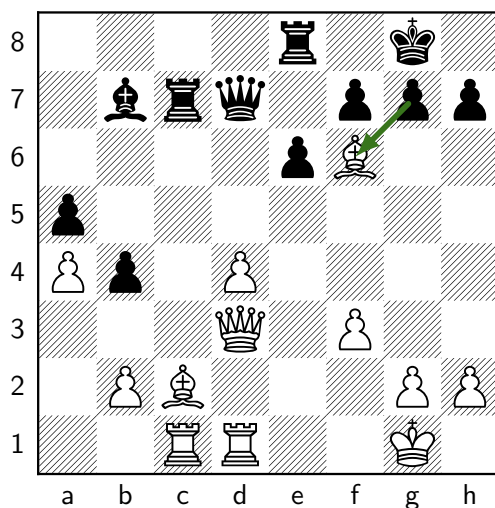
(a)



(b)

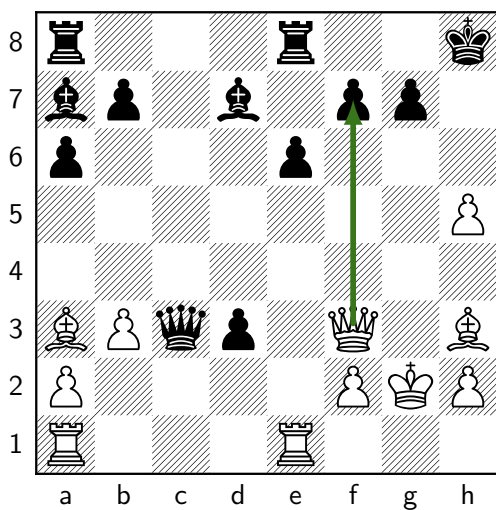


(c)

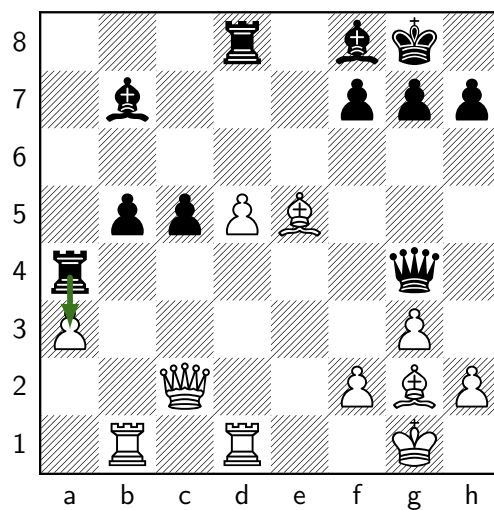


(d)

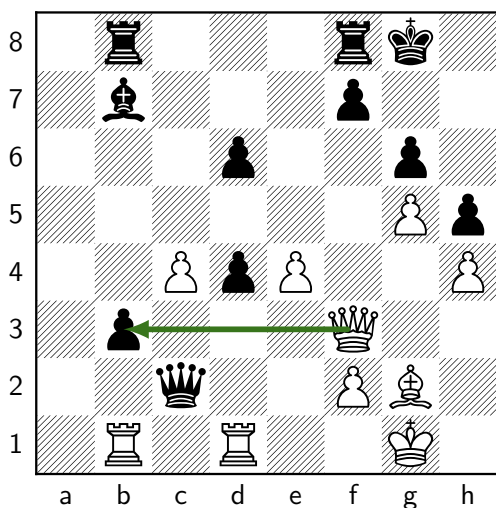
Figure 23: **Move 30, White: dxc3.** (a) Our position; Melanie plays dxc3 (d2c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#496` (ply 66) [structural distance $d=117$, QAP obj=78]: the master played `exd5`; under μ our `Pd2` plays the role of that game’s `Pe4`; cross-type roles: our `Qf3` as their `Rc1`, our `Bh3` as their `Rd1`; transferred move `d2c3` (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3821` (ply 46) [structural distance $d=112$, QAP obj=63]: the master played `Bxc4`; under μ our `Pd2` plays the role of that game’s `Bd3` (a **cross-type** correspondence); cross-type roles: our `Pf2` as their `Rd1`, our `Bh3` as their `Rg1`; transferred move `d2c3` (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2354` (ply 47) [structural distance $d=106$, QAP obj=54]: the master played `gxf6`; under μ our `Pd2` plays the role of that game’s `Pg2`; cross-type roles: our `Ba3` as their `Re8`, our `Ph5` as their `Bb7`; transferred move `d2c3` (**endorses this move**). Synthesis: 4 retrieved analogs transferred to dxc3 (summed QAP objective 229, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



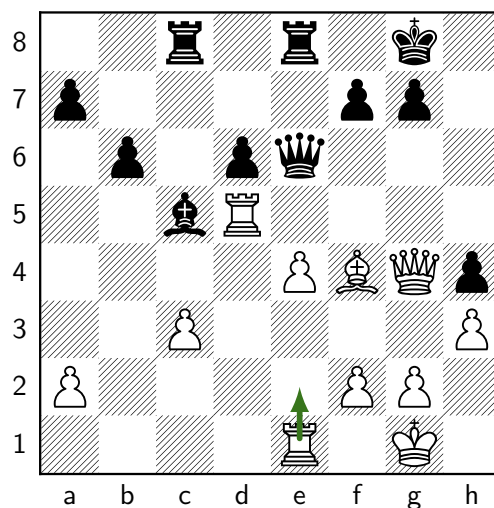
(a)



(b)

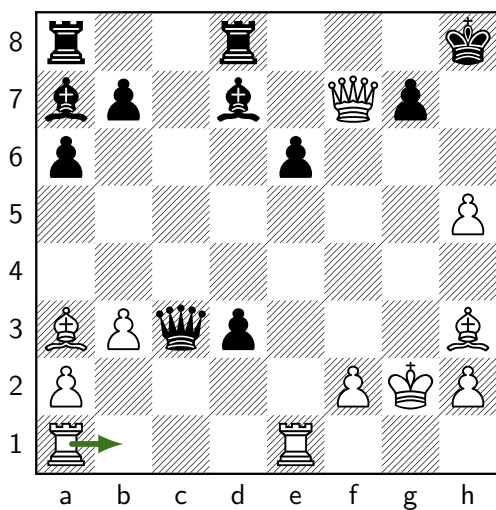


(c)

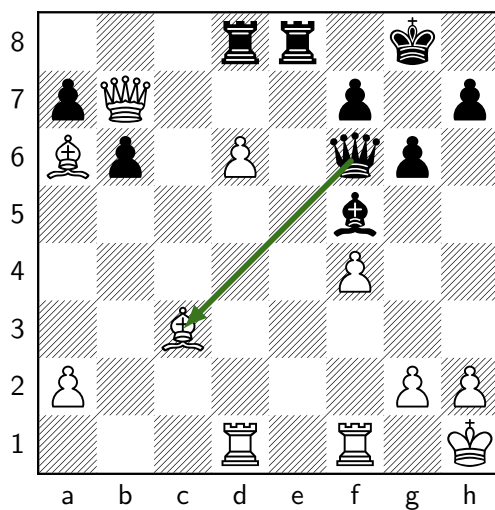


(d)

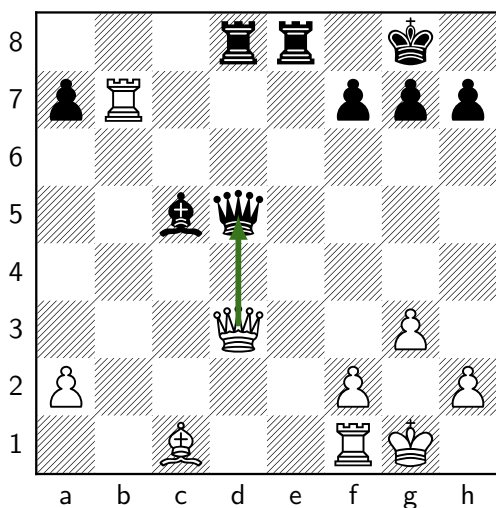
Figure 24: **Move 31, White: Qxf7.** (a) Our position; Melanie plays Qxf7 (f3f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#198` (ply 51) [structural distance $d=95$, QAP obj=93]: the master played Rxa3; under μ our *Re1* plays the role of that game’s *Ra5*; cross-type roles: our *Qf3* as their *Rd8*, our *pf7* as their *Bf8*; transferred move e1e5. (c) Retrieved from `ccr1_top_p0002#1042` (ply 58) [structural distance $d=105$, QAP obj=87]: the master played Qxb3; under μ our *Qf3* plays the role of that game’s *Qf3*; cross-type roles: our *Pf2* as their *Bg2*, our *Bh3* as their *Pg5*; transferred move f3d3. (d) Retrieved from `ccr1_top_p0002#2978` (ply 72) [structural distance $d=101$, QAP obj=70]: the master played Re2; under μ our *Bh3* plays the role of that game’s *Re1* (a **cross-type** correspondence); cross-type roles: our *bd7* as their *Pa2*, our *Ba3* as their *Ph3*; transferred move h3g4. Synthesis: no retrieved analog transferred to Qxf7 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



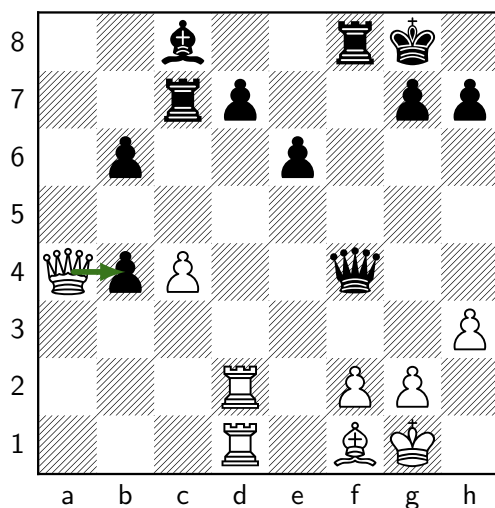
(a)



(b)

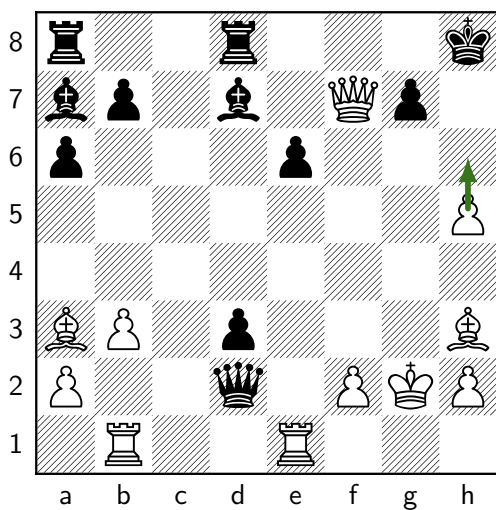


(c)

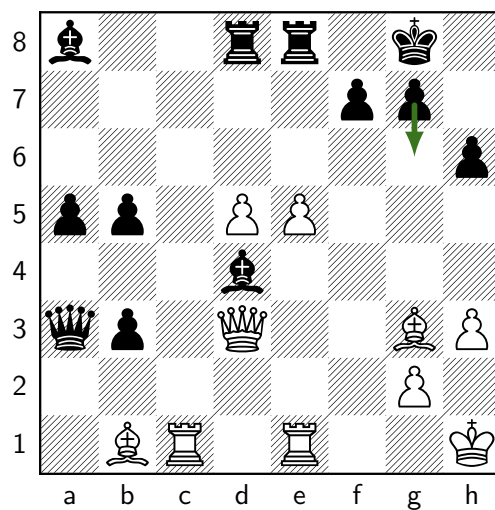


(d)

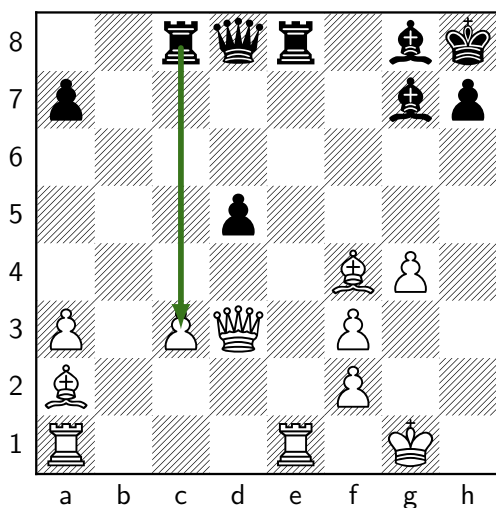
Figure 25: **Move 32, White: Rab1**. (a) Our position; Melanie plays Rab1 (a1b1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2153` (ply 53) [structural distance $d=80$, QAP obj=36]: the master played Qxc3; under μ our *Ra1* plays the role of that game’s *Qf3* (a **cross-type** correspondence); cross-type roles: our *qc3* as their *Rd8*, our *pd3* as their *Re8*; transferred move a1b1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2365` (ply 46) [structural distance $d=70$, QAP obj=34]: the master played Qxd5; under μ our *Ra1* plays the role of that game’s *Qd3* (a **cross-type** correspondence); cross-type roles: our *bd7* as their *Rf1*, our *ba7* as their *Ph2*; transferred move a1b1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3472` (ply 46) [structural distance $d=71$, QAP obj=79]: the master played Qxb4; under μ our *Qf7* plays the role of that game’s *Qa4*; cross-type roles: our *Pa2* as their *Rd1*, our *Ba3* as their *Pg2*; transferred move f7g6. Synthesis: 2 retrieved analogs transferred to Rab1 (summed QAP objective 69, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



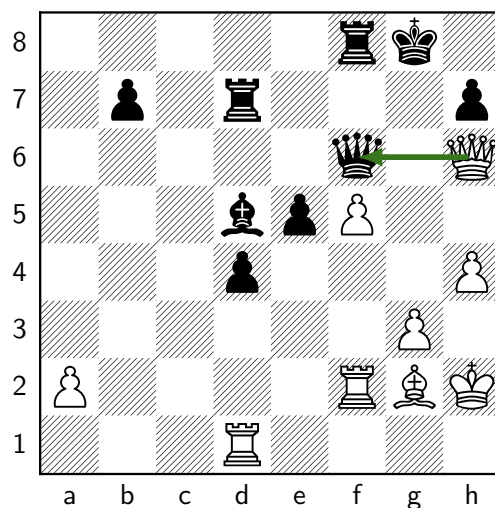
(a)



(b)

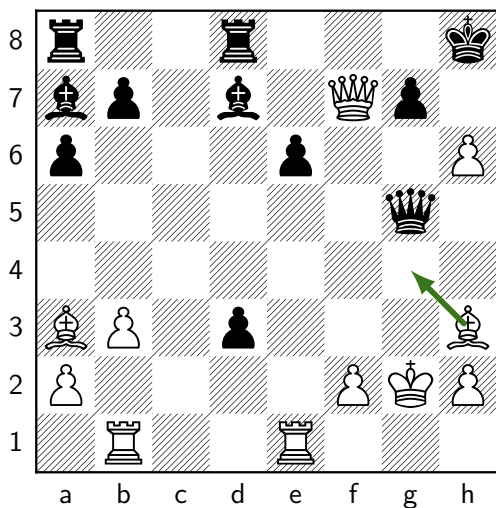


(c)

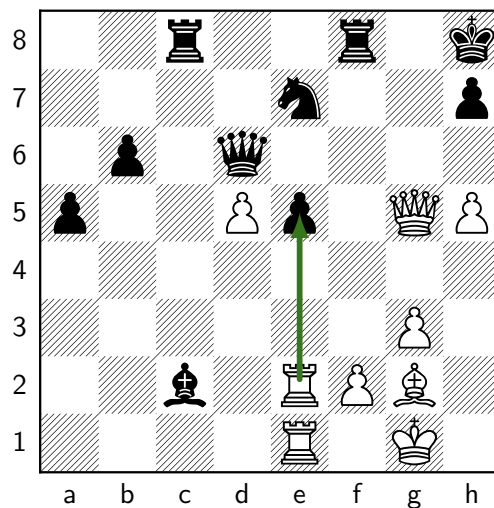


(d)

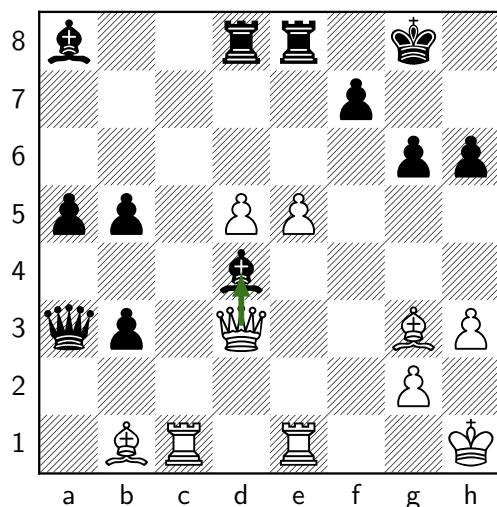
Figure 26: **Move 33, White: h6.** (a) Our position; Melanie plays h6 (h5h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1388` (ply 71) [structural distance $d=60$, QAP obj=78]: the master played g6; under μ our *Ph5* plays the role of that game’s *Pg2*; cross-type roles: our *Bh3* as their *Re8*, our *ra8* as their *Pf7*; transferred move h5h6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1158` (ply 51) [structural distance $d=86$, QAP obj=52]: the master played Rxc3; under μ our *Ph5* plays the role of that game’s *Rc1* (a **cross-type** correspondence); cross-type roles: our *Ba3* as their *Re8*, our *Rb1* as their *Bg8*; transferred move h5h6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2310` (ply 76) [structural distance $d=86$, QAP obj=56]: the master played Qxf6; under μ our *Re1* plays the role of that game’s *Qh6* (a **cross-type** correspondence); cross-type roles: our *Bh3* as their *Rd1*, our *Rb1* as their *Pa2*; transferred move e1e5. Synthesis: 2 retrieved analogs transferred to h6 (summed QAP objective 130, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



(a)

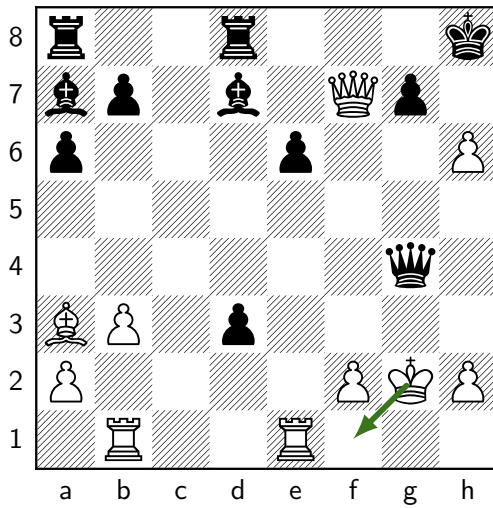


(b)

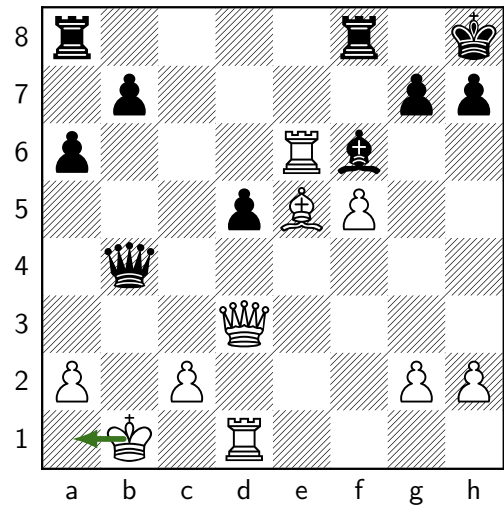


(c)

Figure 27: **Move 34, White: Bg4.** (a) Our position; Melanie plays Bg4 (h3g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1810` (ply 64) [structural distance $d=95$, QAP obj=42]: the master played Rxe5; under μ our Bh3 plays the role of that game’s Re2 (a **cross-type** correspondence); cross-type roles: our Ph2 as their Re1, our Re1 as their Bg2; transferred move h3g4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1388` (ply 72) [structural distance $d=86$, QAP obj=20]: the master played Qxd4; under μ our Bh3 plays the role of that game’s Qd3 (a **cross-type** correspondence); cross-type roles: our qg5 as their Bb1, our bd7 as their Re1; transferred move h3g4 (**endorses this move**). Synthesis: 2 retrieved analogs transferred to Bg4 (summed QAP objective 62, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

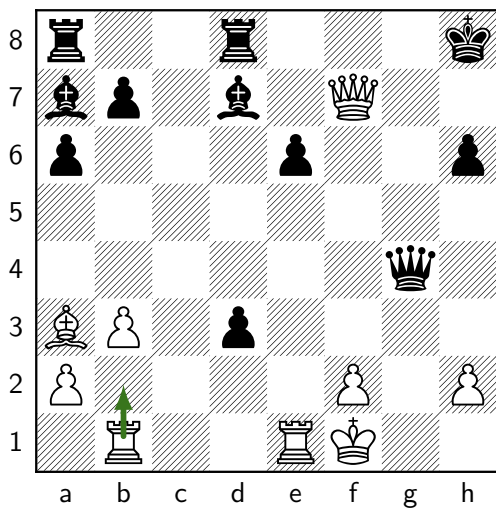


(a)

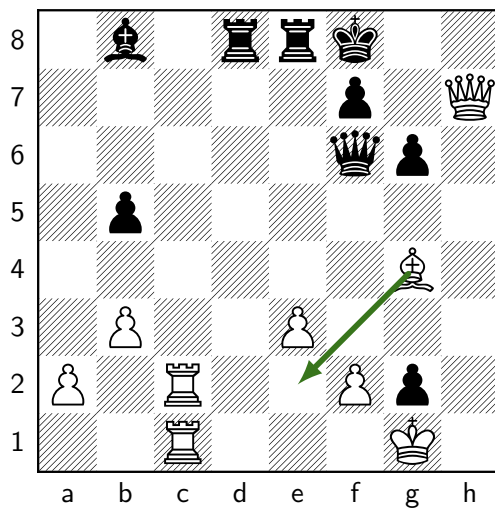


(b)

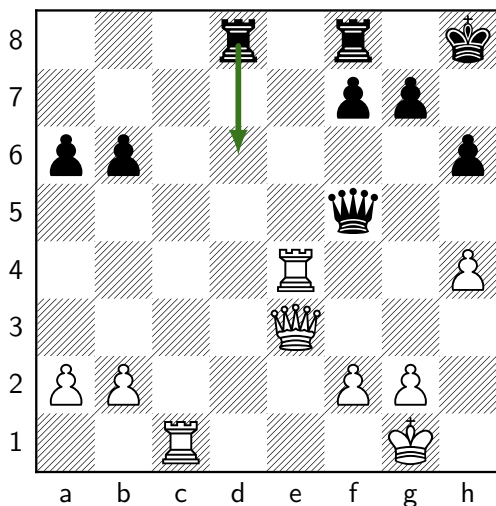
Figure 28: **Move 35, White: Kf1.** (a) Our position; Melanie plays Kf1 (g2f1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3551` (ply 58) [structural distance $d=79$, QAP obj=54]: the master played Ka1; under μ our *Kg2* plays the role of that game’s *Kb1*; cross-type roles: our *Re1* as their *Ph2*, our *pa6* as their *Qd3*; transferred move g2h1. Synthesis: no retrieved analog transferred to Kf1 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



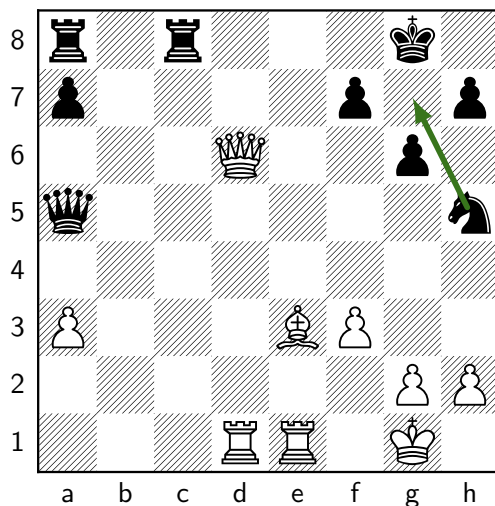
(a)



(b)

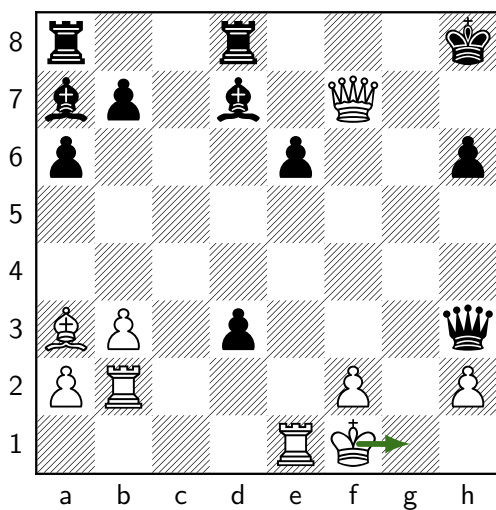


(c)

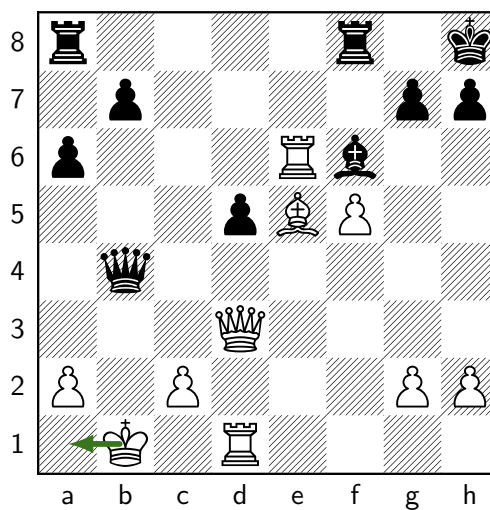


(d)

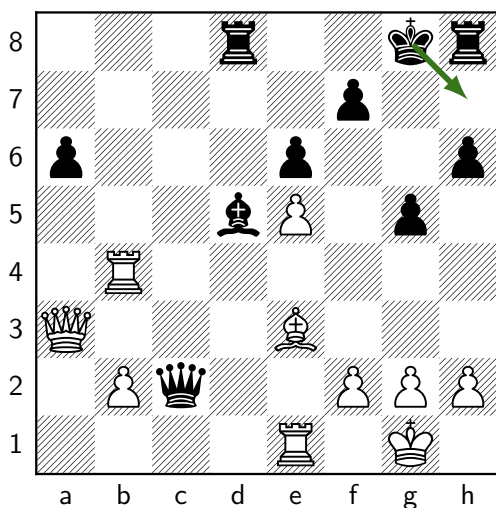
Figure 29: **Move 36, White: Rb2.** (a) Our position; Melanie plays Rb2 (**b1b2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2646` (ply 68) [structural distance $d=79$, QAP obj=30]: the master played Be2; under μ our Rb1 plays the role of that game’s Bg4 (a **cross-type** correspondence); cross-type roles: our qg4 as their Rc2, our Ba3 as their Pb3; transferred move b1b2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#878` (ply 55) [structural distance $d=84$, QAP obj=13]: the master played Rd6; under μ our Rb1 plays the role of that game’s Rd1; cross-type roles: our ba7 as their Ph6, our bd7 as their re4; transferred move b1b2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#554` (ply 59) [structural distance $d=84$, QAP obj=29]: the master played Ng7; under μ our Pb3 plays the role of that game’s Nh4 (a **cross-type** correspondence); cross-type roles: our ba7 as their Ph7, our Ba3 as their Pg6; transferred move b3b4. Synthesis: 2 retrieved analogs transferred to Rb2 (summed QAP objective 43, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



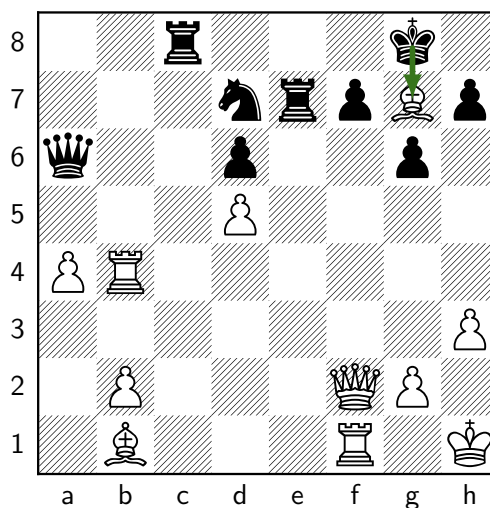
(a)



(b)

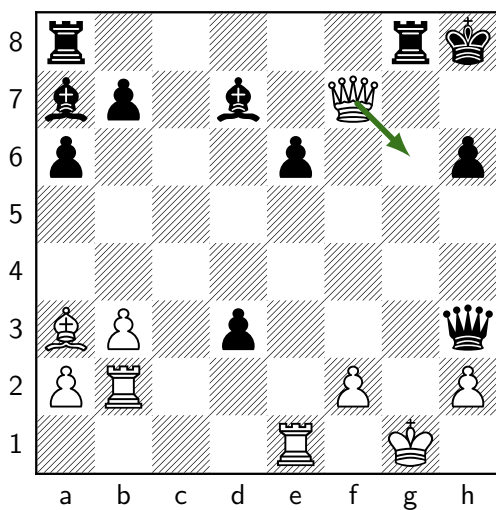


(c)

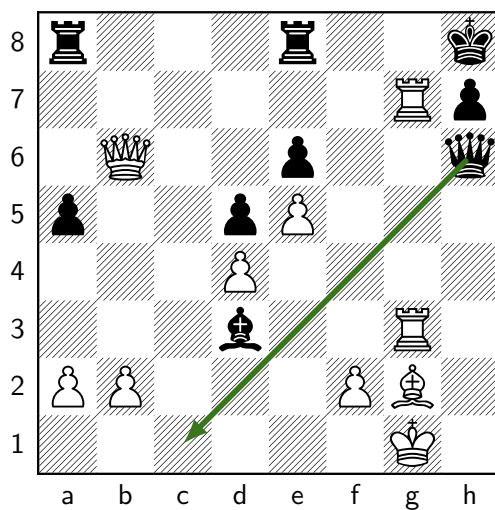


(d)

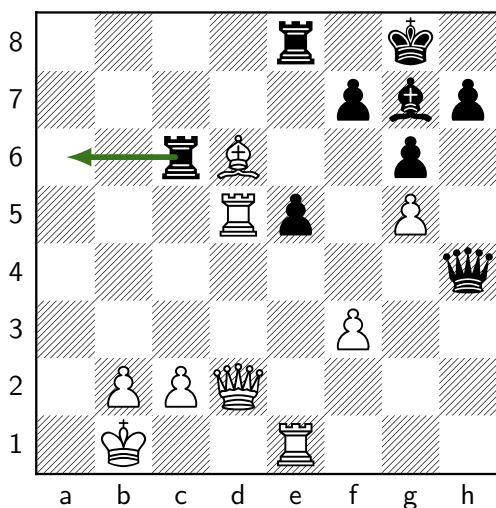
Figure 30: **Move 37, White: Kg1.** (a) Our position; Melanie plays Kg1 (**f1g1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#3551` (ply 58) [structural distance $d=61$, QAP obj=50]: the master played Ka1; under μ our Kf1 plays the role of that game’s Kb1; cross-type roles: our bd7 as their pd5, our pa6 as their Be5; transferred move f1g1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#793` (ply 57) [structural distance $d=68$, QAP obj=4]: the master played Kh7; under μ our Kf1 plays the role of that game’s Kg1; cross-type roles: our Pf2 as their Rh8, our Rb2 as their Pe6; transferred move f1g1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#3742` (ply 61) [structural distance $d=54$, QAP obj=3]: the master played Kxg7; under μ our Kf1 plays the role of that game’s Kg1; cross-type roles: our Pb3 as their Nd7, our Pa2 as their Re7; transferred move f1g1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kg1 (summed QAP objective 58, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



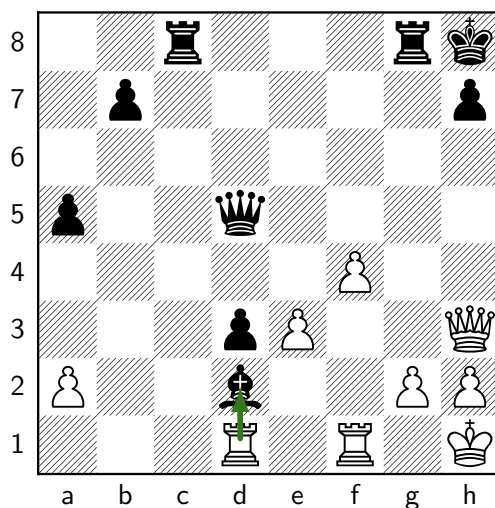
(a)



(b)

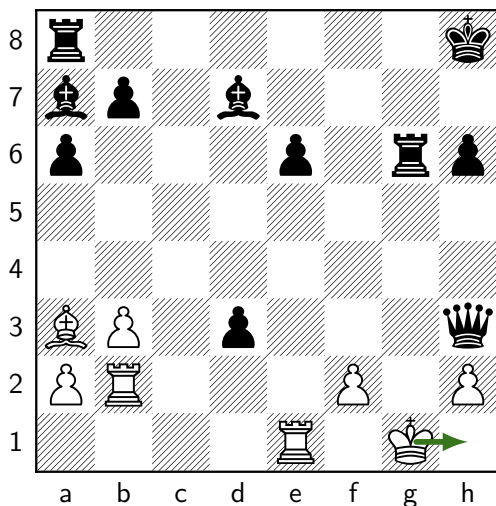


(c)

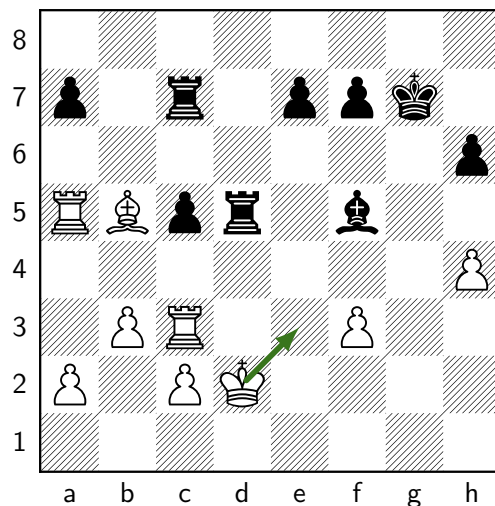


(d)

Figure 31: **Move 38, White: Qg6.** (a) Our position; Melanie plays Qg6 (f7g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2564` (ply 71) [structural distance $d=61$, QAP obj=25]: the master played Qc1+; under μ our Qf7 plays the role of that game's Qh3; cross-type roles: our pd3 as their Ra8, our ba7 as their qb6; transferred move f7g6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3378` (ply 63) [structural distance $d=65$, QAP obj=24]: the master played Ra6; under μ our Qf7 plays the role of that game's Rc3 (a **cross-type** correspondence); cross-type roles: our Rb2 as their Bg7, our rg8 as their bd6; transferred move f7g6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1314` (ply 60) [structural distance $d=62$, QAP obj=22]: the master played Rxd2; under μ our Qf7 plays the role of that game's Rd1 (a **cross-type** correspondence); cross-type roles: our Rb2 as their Pa2, our rg8 as their bd2; transferred move f7g6 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Qg6 (summed QAP objective 153, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



(a)



(b)

Figure 32: **Move 39, White: Kh1.** (a) Our position; Melanie plays Kh1 (g1h1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3406` (ply 64) [structural distance $d=76$, QAP obj=50]: the master played Ke3; under μ our *Kg1* plays the role of that game’s *Kd2*; cross-type roles: our *Re1* as their *Pa2*, our *ra8* as their *Pc2*; transferred move g1h1 (**endorses this move**). Synthesis: 1 retrieved analog transferred to Kh1 (summed QAP objective 50, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.